The Effect of User-Defined Mid-Air Gestures Elicited by On-Screen Visual Properties on Human Biomechanics, Behaviour, and Perception

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Abstract

Evidence shows that visual properties of on-screen objects determine their function and operation, thus influencing gesture elicitation. Therefore, this two-stage experimental study investigated how user-defined mid-air gestures elicited by on-screen visual properties affect human biomechanics, behaviour, and perception. Results showed that, psychologically, for the command ‘next’, swiping with the visual flow provided benefits over swiping perpendicular to it; for ‘remove’, swiping perpendicular to the visual flow offered advantages compared to swiping with it. Behaviourally, for ‘next’, swiping left generated shorter response times than swiping up with a stimulus possessing a horizontal visual flow. Physiologically, for ‘next’, swiping up induced higher forearm muscular load than swiping left under the vertical visual flow condition; for ‘view’, tapping twice resulted in less finger muscular load under the multiple entities condition than the one entity condition. These findings could provide a scientific basis for guiding appropriate mid-air gesture design considering on-screen visual properties.

Keywords

Mid-air interaction; Gesture elicitation; On-screen visual properties; sEMG

1 Introduction

Mid-air interaction is a rapidly evolving field that encompasses innovative methods for human engagement with interfaces or virtual objects without the need for physical contact or traditional input devices. The natural user interface in mid-air interaction can be classified as including voice (use of speech), gaze (use of visual interaction), and gesture (use of body movements) (Fernandez et al., 2016; Ballmer, 2010; Guerino and Valentim, 2020). With recent advancements in computer vision as well as hand and body tracking algorithms (Aigner et al., 2012; Cabreira and Hwang, 2015; Fiorentino et al., 2013, 2016), mid-air gestures, executed without physical contact or traditional input devices, have become a popular input modality, offering users a natural and intuitive way to interact with digital interfaces (Chen et al., 2018; Pereira et al., 2015; Huang et al., 2021).

The rising popularity of mid-air gesture interaction has prompted researchers to focus on optimising gesture-based user interfaces on remote displays, such as interactive tabletops, smart TVs, and interactive kiosks. User interfaces consist of various on-screen objects, such as icons and menus, characterised by distinct visual properties. Several studies have manipulated on-screen objects and their visual properties to create diverse interfaces and explored their impact on user experience in mid-air interaction, particularly during the execution of predefined gestures and tasks, primarily focused on how predefined gestures are performed in response to these visual properties (e.g., Ren and O'Neill, 2012; Mao et al., 2023). However, evidence suggested that specific visual properties of on-screen objects can determine their own function and operation, thereby influencing gesture elicitation in touchscreen interaction (Kim and Lee, 2020). As users' choices for gestures differ significantly between touchscreen and mid-air contexts (Jakobsen et al., 2015), there is a lack of research examining how different visual properties of on-screen objects elicit user-defined mid-air gestures, and how these visual property-elicited mid-air gestures impact the user experience across psychological, behavioural, and physiological aspects.

Therefore, based on the previous studies, this study aims to investigate the effect of user-defined mid-air gestures elicited by on-screen visual properties on user experience. Since mid-air gestures might pose a potential risk of musculoskeletal disorders under long-term use, as even small movements with low forces could increase the risk of injury when repeated frequently (Asakawa et al., 2022; Gerr et al., 2014), we will explore the impact of visual property-elicited gestures on user psychology, behaviour, and physiology by combining the indices of subjective ratings, response times, and sEMG (surface electromyography), respectively. Specifically, we conducted a two-stage experimental study. Experiment 1 used the user-defined approach (Wobbrock et al., 2009) and choice-based elicitation design (e.g., Dim et al., 2016) to obtain the corresponding user-defined mid-air gestures to different visual patterns. Based on the results of experiment 1, experiment 2 conducted a comprehensive evaluation to holistically assess the impact of mid-air gestures on user subjective ratings, response times, and muscle activities (measured by sEMG). Figure 1 shows the workflow, methods, and outcomes of this study. Our study extends beyond traditional task-specific evaluations in gesture-centric research, highlighting the importance of a well-matched visual interface and gesture design in enhancing the overall user experience in mid-air interaction, including in contexts such as remote displays.

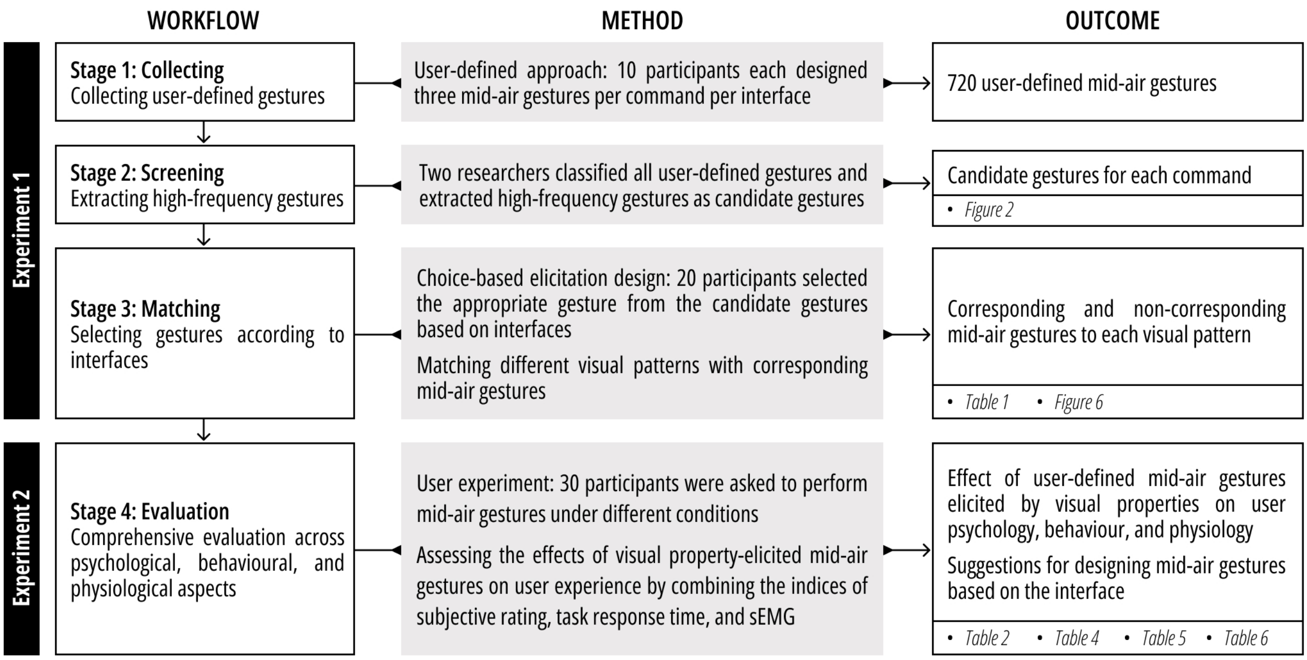

Figure 1. Workflow, methods, and outcomes of this study.

2 Related works

2.1 Visual properties of on-screen objects

An interface comprises various on-screen objects, such as boxes, icons, menus, and pictures (e.g., Issa et al., 2012), each characterised by visual properties such as shape, number of entities, and visual flow. These properties are critical elements of interface design. In terms of shape, a fundamental visual property, it is regarded as 'the nature of an entity' (Ortiz, 2010; Grady, 1997), which interacts with category information throughout the visual hierarchy, fundamentally shaping object perception (Bracci and de Beeck, 2016). In terms of number of entities, it is a property often manipulated in interface-related research, particularly in studies of menu interfaces (e.g., Li et al., 2021; Mao et al., 2024) and affordance on interfaces (e.g., Kim and Lee, 2020). In terms of visual flow, it encompasses path patterns such as straight, curved and central visual flows (Wei et al., 2019). Straight visual flow is the one along the horizontal and vertical directions, with modules arranged from left to right or from top to bottom, commonly categorised as horizontal and vertical layouts in many studies (e.g., Kim and Lee, 2023; Ramón et al., 2016). For user interface design, organising on-screen objects and defining their visual properties are crucial, as interconnected decisions should be made from the selection of functionality to various determinations on how to present them and implement their interaction (Oulasvirta et al., 2020).

In human-computer interaction (HCI), the user interface serves as the visible component of a system through which users interact with computers (Head, 1999), where diverse on-screen visual properties have the potential to influence user perception, behaviour, and physiology. Previous research has examined the impact of visual properties on user experience, often by manipulating different on-screen visual properties to create various user interfaces. First, from the aspect of perception, Zhang et al. (2023) explored the impact of mobile learning platform layout on users' study efficiency and cognitive indicators and found that vertical sequential layout improved efficiency and satisfaction among participants. Second, from the aspect of user behaviour and performance, Ren and O'Neill (2012) evaluated the selection time, error rate and user performance of freehand gestural target selection with different 3D marking menu layouts and target directions, and found that a rectangular layout led to faster execution than an octagonal layout. Third, from the aspect of user physiology, Mao et al. (2023) analysed muscle activities, performance, and user experience to compare panel and pie menus in the context of mid-air interaction within a photo booth, where users performed air tap gestures, and found that the pie menu reduced muscular load, achieved shorter task completion times, and delivered a more enjoyable user experience. Furthermore, Mao et al. (2024) examined the influence of option quantities and layouts (landscape and portrait) on muscle activity, wrist movement, user performance, and subjective assessment when swiping with the index finger and found that different quantities affected efficiency and different combinations of quantities and layouts led to varying physiological load. These studies have explored the impact of on-screen visual properties on user experience through predefined gestures, aiming to determine the optimal visual properties or user interfaces for accomplishing specific tasks.

Nevertheless, research by Kim and Lee (2020) has shown that certain visual properties of on-screen objects can determine their function and operation, thus influencing gesture elicitation in touchscreen interaction. They also found that gestures corresponding to visual flows resulted in faster response times, reflecting an object-based correspondence effect, consequently impacting human behaviours in touchscreen interaction (Kim and Lee, 2023). However, it is important to note that mid-air interaction and touchscreen interaction are distinct modalities, differing significantly in several key aspects, such as the coupling of display and input space, the presence or absence of tactile feedback, the methods used to delimit gestures, which fundamentally influence users' choices of gestures in each modality (Jakobsen et al., 2015). Despite this, how various visual properties impact mid-air gesture elicitation remains largely unexplored, leaving a gap in understanding the effect of visual property-elicited gestures on human perceptual, behavioural, and biomechanical responses.

2.2 Mid-air gesture elicitation

Recently, the natural user interface has been considered a hot topic in HCI (Guerino and Valentim, 2020), which can be classified as multitouch (use of hand gestures in touchscreen), voice (use of speech), gaze (use of visual interaction), and gesture (use of body movements) (Fernandez et al., 2016; Ballmer, 2010; Guerino and Valentim, 2020). With technological advances enabling real-time finger, hand and body tracking (Ardito et al., 2014), mid-air gestures, executed without physical contact or traditional input devices, have become a popular input modality among these modalities, offering users a natural and intuitive way to interact with digital interfaces or virtual objects.

Given the growing popularity and potential of mid-air gestures, designing effective and user-friendly gestures in mid-air interaction is crucial. Gesture elicitation methodologies can be broadly categorised as expert-led methodologies, computationally-based methodologies, and user-led methodologies (Xia et al., 2022). Among these methodologies, user-led methodologies generate gesture vocabularies based on user behaviour or feedback about which gestures most naturally map to a given command (Nielsen et al., 2004; Pyryeskin et al., 2012; Wobbrock et al., 2009). These user-led methodologies are believed to elicit intuitive and discoverable gestures (Xia et al., 2022) and have been widely applied in recent studies. On the one hand, there is research prompting users to select gestures from a set of common gestures, known as the choice-based elicitation study, consisting of a guessability study and a re-examination of the referents that scored low during the previous phase (Koutsabasis and Vogiatzidakis, 2019). The approach was followed in Dim et al. (2016), where users were asked to select their favourite gesture from a predefined list of choices, to investigate mid-air gestures that enable blind people to control the TV. This approach has also been used in touchscreen interaction. For instance, Kim and Lee (2020) asked participants to choose possible gestures from seven common gestures for each function according to various stimuli and found that user preference for gestures varied with the layout and number of images on the screen. Additionally, in a recent study of Kim and Lee (2023), twenty participants were guided to select the most appropriate gesture for combinations of four functions (view, back, remove, and next) and five gestures (tap, double tap, long press, slide, and scroll) across different visual properties, aiming to understand users' mental models on function-gesture mappings in terms of visual patterns. On the other hand, a range of studies directly collected gestures set from end-users, known as the user-defined approach, which is preferred by most participants (Morris et al., 2010). This user-defined approach was first proposed by Wobbrock et al. (2009) to define a set of gestures for surface computing, where they gathered a gesture set by recruiting actual users to design at least one gesture for each given command, and then researchers evaluated the proposed gestures to determine the optimal set. Recently, several studies have utilised the user-defined approach to create user-friendly and intuitive mid-air gestures. For instance, Huang et al. (2021) asked each participant to define two mid-air gestures for each given command without any hints to obtain candidate mid-air gestures for their assessment system for the end-user elicitation experiment. Similarly, to establish a set of hand-based languages available for 3D conceptual design in an MR environment, Sun et al. (2024) summarised 31 fundamental functionalities and collected approximately 930 gesture actions using a user-defined gesture elicitation design with twelve participants. In these studies, the user-defined gestures reflect users' mental models and have shown good performance and high preference (Morris et al., 2010; Wobbrock et al., 2009). Nonetheless, in mid-air interaction, most gesture elicitation studies focus primarily on commands or referents. An exception is the work by Zhou and Bai (2023), which examined the impact of map size on mid-air gesture elicitation and revealed that user-defined gestures for the same geographic information system (GIS) operation commands varied across different map sizes in GIS applications. This highlights a broader issue in that most studies overlook the visual properties of different interface objects when eliciting gestures.

2.3 sEMG applied in mid-air interaction

The sEMG (surface electromyography) is a biomedical signal acquired by the electrical responses generated in muscles throughout its contraction and relaxation, symbolising neuromuscular activities (contraction/relaxation) (Fattah et al., 2012). sEMG is often used to measure muscle activity, indicating ease of execution and ergonomics when assessing gesture interaction.

In recent years, arm fatigue has become an essential factor in the design of human-computer interfaces (Bachynskyi et al., 2015; Hincapié-Ramos et al., 2014). In mid-air interaction, the gestures might pose a potential risk of musculoskeletal disorders under long-term use, with previous studies showing that even small movements with low forces could add to the risk of this injury when repeated frequently (Asakawa et al., 2022; Gerr et al., 2014). Despite being equipment-intensive, requiring specialised skills, time-consuming, and costly (Wang et al., 2018), sEMG has been widely adopted to assess the physiological load on the upper limbs in mid-air gesture interactions due to its ability to provide precise quantitative values for ergonomics, rich biomechanical information, and continuous monitoring of muscle activity. First, sEMG has been directly used to measure muscle activity to assess the human-centred aspects of interaction. For example, Huang et al. (2022a) compared the differences between touchscreen and mid-air interactions when performing swipe gestures on a tablet, combining the indices of sEMG, electrogoniometry, user performance, and subjective assessment, and found that mid-air interaction resulted in lower muscular load, smaller wrist joint excursions, shorter task completion times, and better subjective ratings compared to touchscreen interaction. Similarly, Mao et al. (2023) analysed muscle activity to measure the physiological differences between performing gestures with panel and pie menus within a photo booth setting and showed that the pie menu led to lower muscular load. Second, sEMG is often utilised as one of the parameters in developing evaluation systems to measure interaction performance. For example, Wang et al. (2018) proposed a comfort evaluation model for mid-air gestures based on muscle activity and joint movement data, considering nineteen muscles and seven degrees of freedom in joint movement, to predict gestures that pose a higher risk of fatigue or injury to joints and muscles. Moreover, Huang et al. (2021) introduced an assessment system combining sEMG indices of five muscles and developed an optimal user-defined mid-air gesture set.

3 Experiment 1

Experiment 1 aimed to identify corresponding gestures to different visual patterns by using a choice-based elicitation design. Given that multimedia information browsing, such as image gallery browsing, is a common task applicable to a wide range of activities (Drossis et al., 2013), our experiment was conducted in the context of image gallery browsing, similar to the scenario adopted by Sluÿters et al. (2023). Additionally, referring to Kim and Lee (2023) and Aigner et al. (2012), four fundamental commands (next, select, view, and remove) were used for this experiment. Command 'next' means to go to the next/next group of pictures; command 'select' means to select the image in the centre; command 'view' means to enlarge and view with more detail; command 'remove' means to remove or erase entities.

Following the gesture extraction approach of Sun et al. (2024), a user-defined gesture elicitation design was first conducted to obtain candidate mid-air gestures for the choice-based elicitation design. Ten participants were recruited to design three mid-air gestures according to each visual interface (Figure 4) for each given command without any hints, resulting in 720 user-defined mid-air gestures (10 participants × 3 gestures × 6 interface stimuli × 4 commands). After that, through regular discussions to deter bias and varied interpretations (Wu et al., 2019), two researchers amalgamated similar user-defined mid-air gestures to simplify frequency accumulation. This process involved grouping gestures with minor variations in execution. For example, gestures with the same hand index and motor intent but different speeds and ranges were grouped together. After that, they counted the frequency of gestures and extracted several high-frequency user-defined mid-air gestures for each command as candidates for experiment 1, as shown in Figure 2.

Figure 2. Illustrations of candidate gestures for the commands 'next', 'select', 'view', and 'remove'.

3.1 Method

3.1.1 Participants

Twenty participants (10 females and 10 males) from different majors and professional backgrounds were recruited at Zhejiang University. The participants were 23.80 ($SD = 2.65$) years old on average and all right-handed without any musculoskeletal symptoms. Additionally, all participants were familiar with computer operation and had no visuomotor impairment affecting their interaction with electronic screens.

3.1.2 Materials

Referring to the experimental materials of Kim and Lee (2023) and applications with interfaces displaying images, such as photo galleries, the on-screen visual patterns of the interface stimuli were manipulated in three visual properties, including shape (square and rectangular), visual flow (horizontal and vertical), and number of entities (one and multiple). The illustration of the interface stimulus set for each visual property is shown in Figure 3.

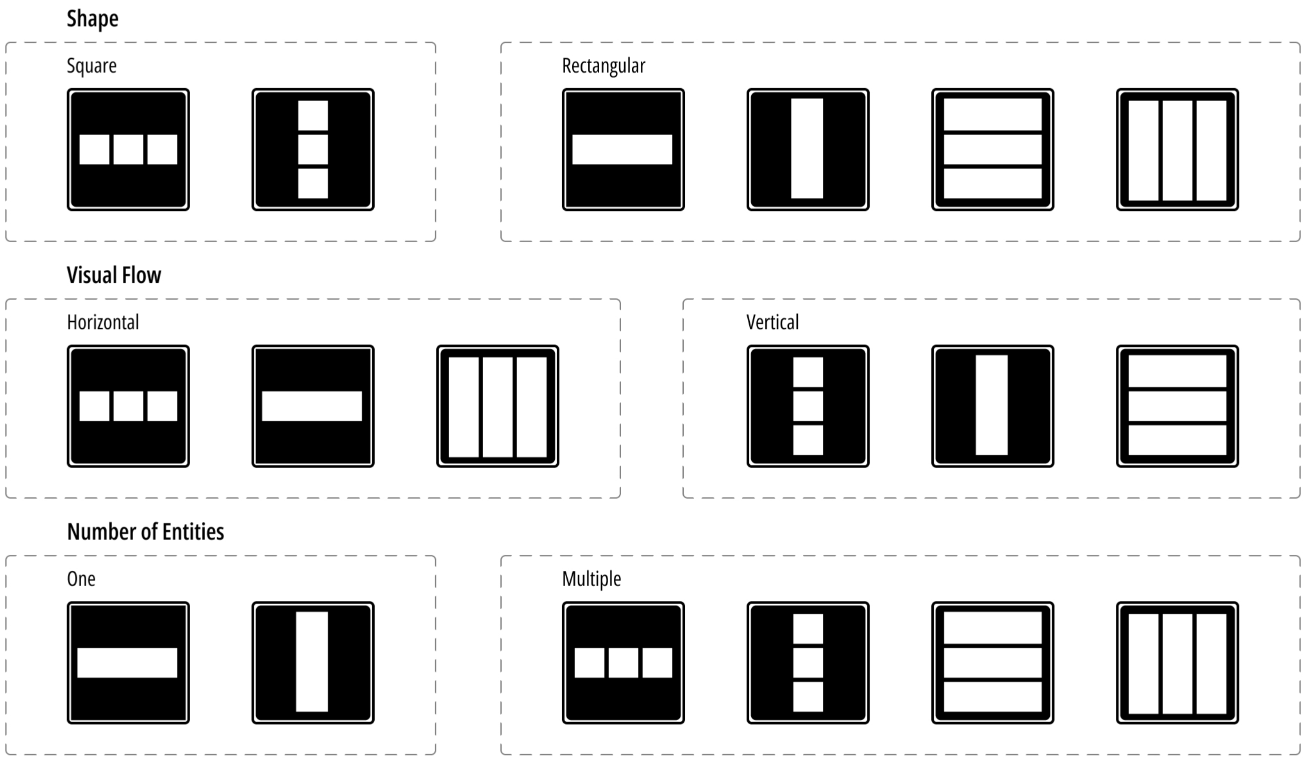

Figure 3. Interface stimulus manipulation in terms of three visual properties: shape (square and rectangular), visual flow (horizontal and vertical), and number of entities (one and multiple).

Based on the above visual patterns, the current study created six square-shaped interface stimuli. The square shape was intended to control for the effect of screen asymmetry between width and height, as suggested by Kim and Lee (2023). All interface stimuli were designed to present images to avoid exogenous variables, i.e., different modalities (e.g., text and video), which users process differently at perceptual and psychological levels (Sundar et al., 2010, 2011). Despite the uniform modality, the stimuli varied in image shape (square and rectangular), visual flow (horizontal and vertical), and number of entities (one and multiple). All images used in the stimuli were randomly selected photographs of natural and urban landscapes all over the world. Please see Figure 4 for the full set of interface stimuli.

Figure 4. Full set of interface stimuli.

3.1.3 Procedure

Experiment 1 was conducted in an ergonomics lab. Participants were seated in an ergonomic chair with adjustable height to ensure comfort and proper posture throughout the experiment. They were instructed to select the most appropriate mid-air gesture for each presented interface stimulus. Instructions and interface stimuli were displayed on an AOC U27U2DP monitor (3840 × 2160, 60 Hz) by slides positioned 90cm from participants' eyes.

Experiment 1 was composed of six experimental trials. At the beginning of the experiment, participants were informed about the procedure. During each trial, an interface stimulus was presented on the right side of the screen for five seconds, followed by candidate mid-air gestures displayed on the left side of the screen for one of the commands. Participants were instructed to select the most appropriate mid-air gesture for executing the command with the current interface stimulus. Then, candidate mid-air gestures for another command were displayed, replacing the previous ones. This process was repeated four times for each of the four commands ('next', 'select', 'view', and 'remove'). The four commands and six trials were randomly presented to each participant to avoid the influence of the experiment sequence. As a result, this experiment eventually collected 24 responses (six interface stimuli on the four commands) from each participant, a total of 480 responses from 20 participants. The experimental schedule is illustrated in Figure 5.

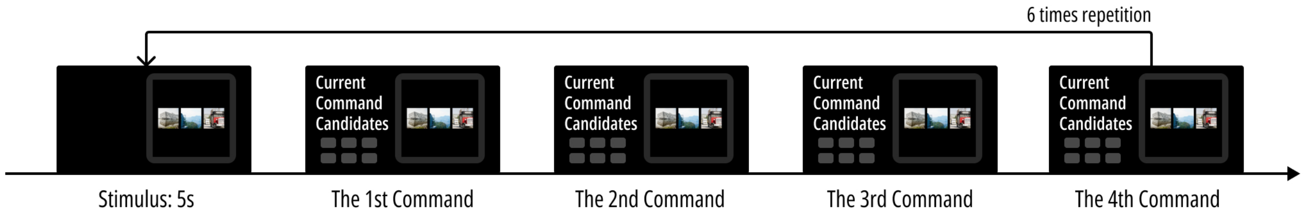

Figure 5. Schedule of experiment 1.

3.2 Results

We conducted a Chi-square test to compare mid-air gestures with each visual property (visual flow, number of entities, and shape) by SPSS software (version 23.0, International Business Machines Corporation, Armonk, NY, USA). For more reliable analysis, rarely selected mid-air gestures were eliminated as outliers. Cramér's V was estimated as the measure of effect size for the Chi-square test.

For the command 'next', the most frequently selected gesture was swiping left with the index finger (44.17%). Others were swiping up with the index finger (26.67%), swiping left (19.17%) /up (9.17%) with the palm, and tapping on the right side with the index finger (0.83%). The rarely selected gesture, tapping on the right side with the index finger, was excluded from the analysis. The Chi-square test revealed a significant difference in gesture selection between the horizontal and vertical visual flow conditions [$\chi^2(3, N = 119) = 28.784, p < 0.001$, Cramér's V=0.492]. Specifically, under the horizontal visual flow condition, swiping left with the index finger was significantly more frequently selected than swiping up with the index finger and swiping up with the palm. Under the vertical visual flow condition, swiping up with the index finger and swiping up with the palm were significantly more frequently selected than swiping left with the index finger. Neither the number of entities (Fisher's exact test, $p = 0.056$, Cramér's V=0.255) nor the shape (Fisher's exact test, $p = 1.000$, Cramér's V=0.036) demonstrated a significant effect on gesture selection.

For the command 'select', the most frequently selected gesture was tapping with the index finger (55.83%). Other gestures included tapping twice with the index finger (20%), palm clenched into a fist (15%), and pinching with the thumb and the index finger (9.17%). The analysis revealed no significant effect of visual flow [$\chi^2(3, N = 120) = 0.495, p = 0.920$, Cramér's V=0.064], number of entities (Fisher's exact test, $p = 0.386$, Cramér's V=0.159), and shape (Fisher's exact test, $p = 0.855$, Cramér's V=0.096).

For the command 'view', the most frequently selected gesture was stretching with the thumb and the index finger (77.5%). Others were tapping twice with the index finger (21.67%) and tapping twice with a fist (0.83%). The rarely selected gesture, tapping twice with a fist, was eliminated before the analysis. There was a significant difference in gesture selection between the one entity and multiple entities conditions [$\chi^2(1, N = 119) = 6.809, p = 0.009$, Cramér's V=0.239]. Under the one entity condition, stretching with the thumb and the index finger was significantly more frequently selected than tapping twice with the index finger, and under the multiple entities condition, tapping twice with the index finger was significantly more frequently selected than stretching with the thumb and the index finger. Additionally, there was a significant difference in gesture selection between the square and rectangular conditions [$\chi^2(1, N = 119) = 4.003, p = 0.045$, Cramér's V=0.183]. Specifically, tapping twice with the index finger was significantly more frequently selected on stimuli with a square shape; stretching with the thumb and the index finger was significantly more frequently selected on stimuli with a rectangular shape. No significant effect of visual flow on gesture selection [$\chi^2(1, N = 119) = 0.242, p = 0.623$, Cramér's V=0.045] was found.

For the command 'remove', the majority of the selection was swiping left with the index finger (36.67%) and swiping up with the index finger (36.67%). Others were drawing a cross with the index finger (10.83%), pinching with the thumb and the index finger (6.67%), swiping left (3.33%)/up (2.5%) with the palm, and slicing diagonally with the palm (3.33%). The rarely selected gestures, swiping left/up with the palm and slicing diagonally with the palm, were eliminated before the analysis. The Chi-square test revealed a significant difference in gesture selection between the horizontal and vertical visual flow conditions (Fisher's exact test, $p < 0.001$, Cramér's V=0.602). Under the horizontal visual flow condition, swiping up with the index finger was significantly more frequently selected than swiping left with the index finger and pinching with the thumb and the index finger; under the vertical visual flow condition, swiping left with the index finger was significantly more frequently selected than swiping up with the index finger and drawing a cross with the index finger. Neither the number of entities (Fisher's exact test, $p = 0.153$, Cramér's V=0.216) nor the shape (Fisher's exact test, $p = 0.866$, Cramér's V=0.091) demonstrated a significant effect on gesture selection.

3.3 Discussion

Experiment 1 identified optimal mid-air gestures corresponding to various visual patterns by investigating the influence of different visual properties on gesture selection.

For the command 'next', swiping left with the index finger was more frequently selected when stimuli featured horizontal flow; swiping up with the index finger and swiping up with the palm was more frequently selected in the presence of vertical visual flow. These results were in line with Kim and Lee (2023) that there was a correspondence between the direction of the gesture and the visual flow of on-screen entities for the function 'next', which might be attributed to the fact that participants could perceive the visual flow of the digital entities as one group, to be manipulated together due to the Gestalt principles of continuity and common fate (Wagemans et al., 2012; Wertheimer, 1923).

For the command 'select', no significant effect on gesture selection was observed across visual properties, including visual flow, number of entities, and shape. These results were plausible because when interacting with graphical user interfaces, selections were almost always made by simple cursor clicks, regardless of on-screen visual properties. Due to the legacy bias (Morris et al., 2014), in mid-air interactions, participants might tend to choose simple traditional one-click gestures, ignoring differences in visual properties.

For the command 'view', in terms of the number of entities, under the one entity condition, stretching with the thumb and the index finger was more frequently selected than tapping twice with the index finger. As 'view' was interpreted as opening and magnifying in our experiment, this result was similar to Sun et al. (2024) and Sluÿters et al. (2023), where participants also stretched with the thumb and the index finger to zoom in and magnify one object. Under the multiple entities condition, tapping twice with the index finger was more frequently selected than stretching with the thumb and the index finger. Considering that interfaces with multiple entities are more visually complex than those with only one entity, and that inaccurate gestures might cause misoperation by jitter and minor instabilities (Drossis et al., 2013), tapping twice with the index finger could be more suitable for interfaces with multiple entities as it is more precise and conveys a sense of double confirmation. In terms of shape, tapping twice with the index finger was more frequently selected on the stimuli with square entities, while stretching with the thumb and the index finger was more frequently selected on the stimuli with rectangular entities. These results could be attributed to the composition of our interface stimuli, that the stimuli with square shapes consistently featured multiple entities, and tapping twice with the index finger was most frequently selected with multiple entities.

For the command 'remove', we found that swiping up with the index finger was more frequently selected on the stimuli with horizontal visual flow; swiping left with the index finger was more frequently selected on the stimuli with vertical visual flow. These results might be attributed to the legacy bias (Morris et al., 2014), as the preferred directions for swiping gestures for the 'remove' command were typically perpendicular to the visual flow directions. For instance, on smartphones, vertically arranged messages in the notification bar were usually removed using a left swipe (Lin et al., 2021), while horizontally arranged background running apps could be removed using an upward swipe according to most cases in life.

Experiment 1 analysed the effect of visual properties (shape, visual flow, and number of entities) on user-defined mid-air gesture selection with four commands ('next', 'select', 'view', and 'remove') through the choice-based gesture elicitation design, and summarised the corresponding gestures to certain visual patterns based on selected frequencies and Chi-square analysis. For the command 'next', the corresponding gesture to the horizontal visual flow was swiping left with the index finger, and to the vertical visual flow was swiping up with the index finger; for the command 'view', the corresponding gesture to one entity was stretching with the thumb and the index finger, and to multiple entities was tapping twice with the index finger; for the command 'remove', the corresponding gesture to the horizontal visual flow was swiping up with the index finger, and to the vertical visual flow was swiping left with the index finger. Corresponding gestures to different visual patterns are illustrated in Figure 6. These mid-air gestures differed from those obtained by Kim and Lee (2023) in touchscreen research, partly due to our approach of allowing users first to define mid-air gestures and then select from them, rather than using a closed elicitation method that restricts participants to predefined gestures (Villarreal-Narvaez et al., 2020). This method likely provided a more accurate reflection of users' preferences. Additionally, these results further confirmed that mid-air and touchscreen interactions are fundamentally distinct modalities, impacting users' choice of gestures.

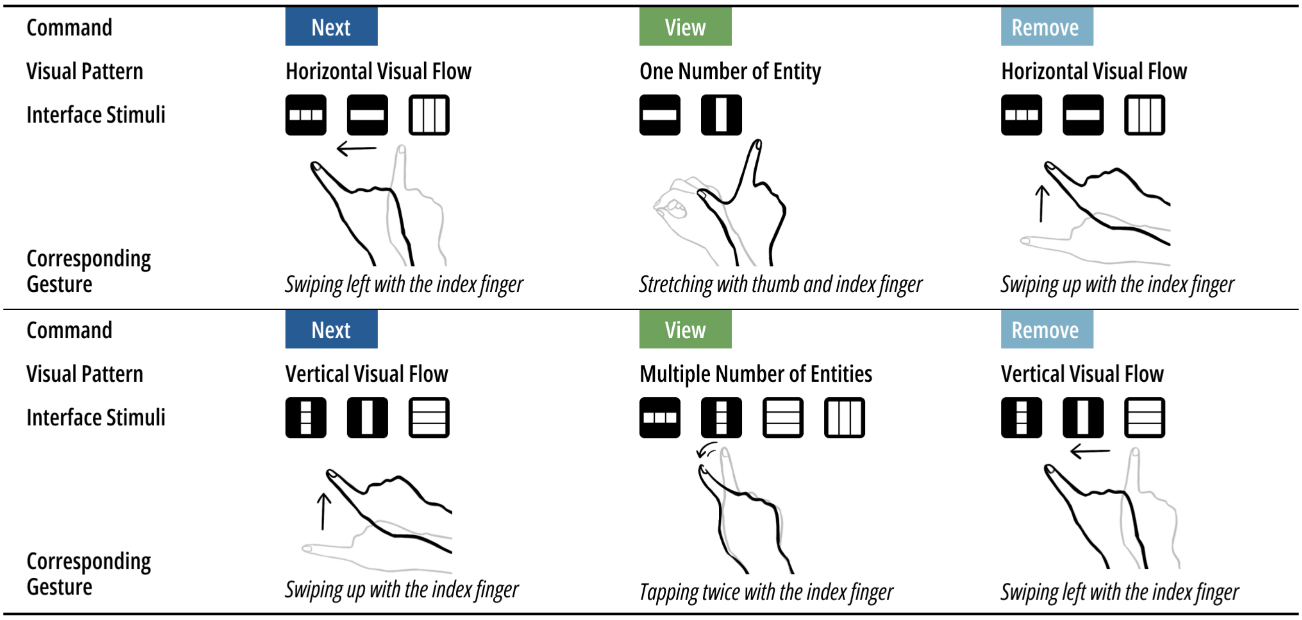

Figure 6. The corresponding gestures to different visual patterns.

However, the impact of these mid-air gestures elicited by on-screen visual properties on user experience was unknown, especially from psychological and physiological aspects. Given that some results of experiment 1 might have been influenced by bias, experiment 2 was structured to evaluate these visual property-elicited gestures from an ergonomic perspective for the commands 'next', 'view', and 'remove', which showed significant results in experiment 1.

4 Experiment 2

The aim of experiment 2 was to evaluate the effect of user-defined mid-air gestures elicited by visual properties on user experience from psychological, behavioural and physiological aspects. Based on the results of experiment 1, for the commands 'next', 'view', and 'remove', corresponding gestures (the most preferred user-defined mid-air gestures) and non-corresponding gestures (worst or less preferred mid-air gestures) were defined based on different visual patterns. The selection of the corresponding and non-corresponding gestures was referenced from Kim and Lee (2023). These groupings are outlined in Table 1. The effect of these mid-air gestures on user experience was compared under different visual pattern conditions.

Command	Visual Property	Visual Pattern	Corresponding Gesture	Non-corresponding Gesture
Next	Visual flow	Horizontal Vertical	Swiping left with the index finger Swiping up with the index finger	Swiping up with the index finger Swiping left with the index finger
View	Number of entities	One Multiple	Stretching with the thumb and the index finger Tapping twice with the index finger	Tapping twice with the index finger Stretching with the thumb and the index finger
Remove	Visual flow	Horizontal Vertical	Swiping up with the index finger Swiping left with the index finger	Swiping left with the index finger Swiping up with the index finger

Table 1. Corresponding and non-corresponding gestures to each visual pattern.

We employed a comprehensive approach to evaluate the effect of user-defined mid-air gestures elicited by visual properties on user psychology, behaviour and physiology. Specifically, subjective evaluations were used to assess psychological factors, response time measurements were employed to capture behavioural aspects, and muscle activity measurements were used to reflect physiological responses, which has been widely applied to evaluate physiological loads on upper limbs with various human-computer interaction modes (e.g., Coppola et al., 2018; Huang et al., 2022b; Mao et al., 2023).

4.1 Method

4.1.1 Participants

This experiment was performed by 30 participants (16 females and 14 males) with ages varying between 21 and 29 years old ($M = 22.87, SD = 1.91$), recruited from Zhejiang University. The participants were all right-handed without musculoskeletal symptoms. Besides, they were familiar with computer operations and had no visuomotor impairment affecting interaction with electronic screens. This study was approved by the university office of research ethics. Each participant confirmed the informed consent and answered a questionnaire on demographic information before beginning the experiment.

4.1.2 Materials

Experiment 2 comprised two kinds of stimuli: interface stimuli and gesture stimuli. The set of interface stimuli was the same as experiment 1, consisting of six interfaces, as depicted in Figure 4. The set of gesture stimuli consisted of four semi-transparent videos: (a) a right hand swiping left with the index finger, (b) a right hand swiping up with the index finger, (c) a right hand tapping twice with the index finger, and (d) a right hand stretching with the thumb and the index finger. These mid-air gestures were the corresponding or non-corresponding gestures to each visual pattern, obtained from experiment 1. Therefore, for each of the commands 'next', 'view', and 'remove', this experiment used a 2 (visual pattern: horizontal and vertical/one and multiple) × 2 (mid-air gesture: corresponding gesture and non-corresponding gesture) design. The visual pattern and mid-air gesture factors were within subjects and within items. These interface stimuli and gesture stimuli were edited and merged into one video file in accordance with the below experimental procedure in Adobe After Effects 2022, rendered at a resolution of 1920 × 1080 pixels at 30 frames per second.

4.1.3 Dependent parameters

Psychological (intuitiveness, learnability, complexity, comfort, and efficiency), behavioural (task response time), and physiological (muscle activity) factors were measured to comprehensively evaluate the effect of mid-air gestures elicited by visual properties on user experience.

Psychological parameters

From the psychological aspect, all participants responded to five questions about subjective intuitiveness, learnability, complexity, comfort, and efficiency. The responses were marked on a 10cm visual analogue scale (VAS), with 0 representing the highest level of intuitiveness, learnability, comfort, and efficiency, and the lowest level of complexity (Coppola et al., 2018).

According to Vogiatzidakis and Koutsabasis (2018), these metrics can be categorised into two general groups: those related to the mental model of users (intuitiveness, learnability) and those associated with the ergonomic features of gestures (complexity, comfort, efficiency). Intuitiveness refers to the degree to which a gesture makes use of transferable and existing knowledge (Raskin, 1994), ensuring that the interface design aligns with users' expectations for input (Lao et al., 2009). Learnability refers to the ease with which users can begin effective interaction and achieve optimal performance with a gesture (Dix, 2004). Complexity refers to the degree of difficulty to the movements that must be remembered or performed, also referred to as a mental load, performability, simplicity, or attention (Xia et al., 2022). Comfort refers to the ergonomic aspect of gestures, ensuring that users feel physically well during gesture execution without encountering pain, fatigue, or discomfort (Xia et al., 2022). Efficiency is referred to as the difficulty, human performance, interaction cost, speed, duration, effort, ease of performance, or ease of operation of a gesture action (Xia et al., 2022).

Before the experiment started, participants received explanations regarding the concepts of intuitiveness, learnability, complexity, comfort, and efficiency. After each gesture task trial, participants were asked to respond to the VAS immediately to record their subjective perception. Tick mark distances from the left side of the 10cm line were measured and recorded as the ratings of each item.

Behavioural parameters

From the behavioural aspect, participants were asked to perform mid-air gestures according to the instructions in experiment 2, and their reaction time in each trial was recorded as task response time. This metric was measured to assess the influence of visual property-elicited mid-air gestures on human behaviour. Previous studies have demonstrated the presence of digital affordance on the screen in touchscreen interaction, influencing the perceptuomotor process of human subjects (Kim and Lee, 2023). Given that perceiving affordance is processed behaviorally or neurologically within approximately 250ms (Bub et al., 2021; Chong and Proctor, 2020), when participants need to respond immediately upon seeing stimuli, their reaction time for performing gestures might be affected.

Task response time was recorded by PsychoPy (version 2023.2.1), which was used to present stimuli and collect responses in experiment 2. During the experiment, participants were prompted by a beep sound to perform the mid-air gesture, and task response times were collected for 1500ms from the beginning moment of the beep sound.

Physiological parameters
From the physiological aspect, muscle activities of the anterior deltoid (AD), biceps brachii (BB), flexor carpi ulnaris (FCU), and first dorsal interosseus (FDI) were recorded by Biopac MP150 (HLT100C high-level transducer module, Biopac system, CA, USA). These muscles included the main motor muscles, respectively controlling the shoulder (AD), elbow (BB), wrist (FCU), and index finger (FDI) movements. EMG signals were digitally sampled at 1000 Hz. Surface electrodes were positioned following the guidelines provided by Barbero et al. (2012). Figure 7 illustrates the measured muscles using the BioDigital Human platform (Qualter et al., 2012). Before the experiment, participants were asked to perform the muscle-specific maximal voluntary isometric contractions (MVICs) to determine the maximal voluntary excitation (MVE) for tested muscles, following the protocols outlined by Boettcher et al. (2008) and Forman et al. (2019). During the MVIC trial in each muscle, participants were instructed to ramp up gradually while the experimenter resisted participants' force exertions using up to their entire strength to get an MVIC. Participants were measured by MVIC three times for each muscle and rested for 30s between exertions (Janssen et al., 2015; Holmes et al., 2015).

The original EMG signal was filtered by the digital bandpass filter (10–500 Hz), and full-wave rectification (30ms moving window), and then a 2nd order Butterworth low pass filter was applied ($f_c = 3$ Hz) (Alenabi et al., 2019). The maximum voluntary electrical activity (MVE) of each muscle was defined as the maximum voltage value of the processed signal in MVIC trials (Schwartz et al., 2017). The normalisation of EMG was calculated as the %MVE of each muscle, which was a way that eliminated differences between participants by ensuring the use of absolute values in the comparative analysis. The mean amplitude level in %MVE was used to evaluate the difference in muscular load. In the analysis, the first and last mid-air gesture movements were discarded to mitigate potential behavioural adjustments and anticipatory responses (Gustafsson et al., 2018).

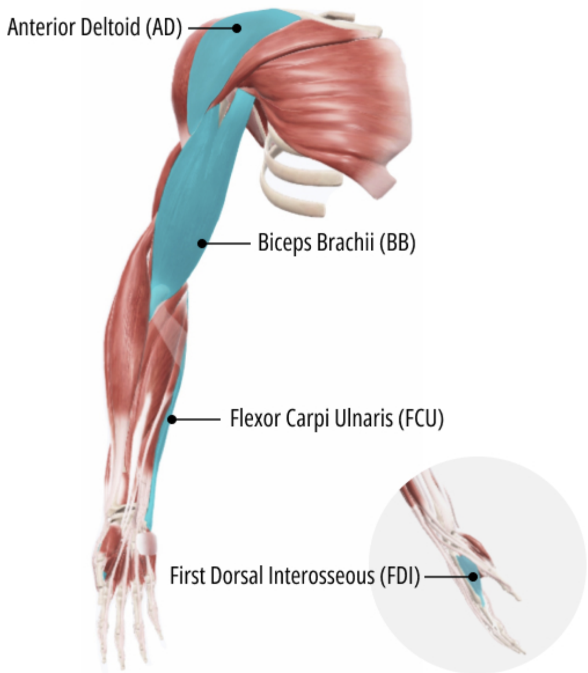

Figure 7. Illustration of four measured muscles of the upper limb.

4.1.4 Procedure

Experiment 2 was a laboratory experiment where participants were asked to sit in an ergonomic chair in front of the AOC U27U2DP monitor (3840 × 2160, 60 Hz) and follow the instructions on the screen to complete all gesture trials. Muscle activities were recorded by Biopac MP150. Task response times were collected when participants released the right key on the keyboard. The presentation of stimuli and collection of response times were controlled by PsychoPy (version 2023.2.1). The monitor height was approximately adjusted to align with both the centre of the fixation cross and the eye level of the participants. The distance from the monitor to the participants' eyes was consistently set at approximately 90cm.

Considering the three commands ('next', 'view', and 'remove'), experiment 2 consisted of three blocks, each focusing on one command. Each block contained 12 trials, with each trial presenting one of the 12 combinations of interface stimuli and mid-air gestures (six interface stimuli and two mid-air gestures for each command). Prior to the experimental trials, participants completed three practice trials to familiarise themselves with the task. During each experimental trial, once the participant pressed the right key, the command would be presented on the screen for three seconds, followed by a fixation cross for 500ms and a blank screen for 500ms. Then, an interface stimulus was presented in the centre of the screen for eight frames (about 267ms), followed by a moving semi-transparent hand gesture stimulus overlapped on the interface stimulus for eight frames (about 267ms). Once the gesture stimulus disappeared, a beep sound was played, and the participant was asked to say the command word while simultaneously lifting their hand from the right key to mimic the mid-air gesture toward the interface stimulus as soon as possible. After that, the participant was required to repeat the same mid-air gesture four times at two-second intervals. The response times for the first execution of the mid-air gesture were recorded, and sEMG data for all five executions were collected. At the end of each trial, participants were asked to provide a subjective assessment by rating the task on a VAS across intuitiveness, learnability, complexity, comfort, and efficiency. The interval between trials was set at one minute, and between blocks was set at five minutes. Within each block, the six interface stimuli and two gesture stimuli combinations were presented in random order. The order of the blocks was counterbalanced to avoid the influence of the experiment sequence. Please see Figure 8 for a schematic experimental procedure within a block. In the experiment, participants were instructed to mimic and perform the mid-air gesture accurately. A gesture recognition model was trained using Create ML (Apple Create ML. https://developer.apple.com/machine-learning/create-ml/) and mid-air gestures that were not successfully recognised by the model were identified as execution failures and would be removed from the analysis.

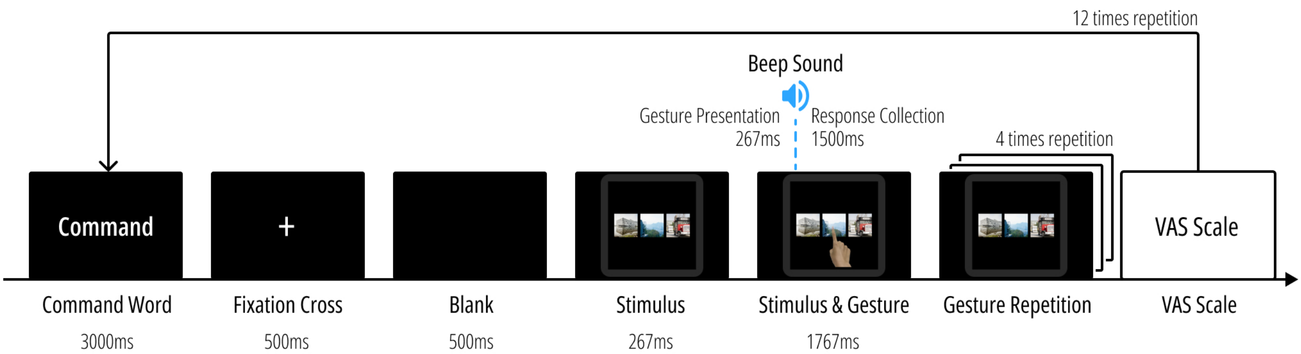

Figure 8. Schematic of experimental procedure within a block of experiment 2.

4.1.5 Data processing and statistics

For the three commands 'next', 'view', and 'remove', a two-way repeated measures ANOVA (visual patterns × mid-air gestures) was carried out for each of the ten dependent variables, including the subjective assessment (intuitiveness, learnability, complexity, comfort, and efficiency), performance (task response time), and sEMG of four measured muscles (AD, BB, FCU, and FDI) by SPSS software (version 22.0, International Business Machines Corporation, Armonk, NY, USA). Bonferroni post hoc tests were used to identify significant differences between visual properties and mid-air gestures. The level of statistical significance was set at 0.05. In addition, for the 'view' command, data obtained in the square stimulus condition (*stimulus (a)* and *stimulus (b)*) were removed from our analysis to prevent interference from the shape factor.

To ensure the reliability and accuracy of our analysis, we employed machine learning for mid-air gesture recognition in experiment 2. This allowed us to verify whether the participants performed the gestures as instructed and to ensure that the executed gestures matched the intended ones. Specifically, we trained a hand action classifier model using Create ML by Apple (Apple Create ML. https://developer.apple.com/machine-learning/create-ml/), based on the graph convolutional neural network algorithm. This model was trained on a custom dataset of 300 video samples, each recorded at a resolution of 1920 × 1080 pixels and a frame rate of 30 fps, using the built-in camera of a MacBook Air 2022. The dataset was designed to capture a diverse range of hand gestures under various conditions, including different lighting and gesture variations. The model was trained to recognise four dynamic hand gestures (swiping left with the index finger, swiping up with the index finger, stretching with the thumb and index finger, and tapping twice with the index finger) and one negative gesture class (irrelevant actions). The training process included 80 iterations with a 30-frame prediction window. The F1 scores were measured as 0.954 for swiping left with the index finger, 0.947 for swiping up with the index finger, 0.966 for stretching with the thumb and index finger, and 0.974 for tapping twice with the index finger. In experiment 2, gesture detection was performed using the built-in camera of a MacBook Air 2022, positioned underneath the stimulus display at a 15-degree upward angle to capture the participant's hand movements. The camera was set at an approximate 15-degree upward angle, focusing on the participant's hand. This setup ensured consistency with the conditions used for the training data.

4.2 Results

Descriptive and statistical results of muscle activities, performance, and subjective ratings are presented in Table 2. Please see Table 4, Table 5, and Table 6 in the *Appendix* for the detailed statistical results of the two-way repeated measures ANOVA for the commands 'next', 'view', and 'remove', respectively.

		Muscle Activity (%MVE)				Performance	Subjective Assessment				
Next		AD	BB^{c}	FCU^{a}	FDI^{c}	Rt	Intuitiveness^{abc}	Learnability^{a}	Complexity^{a}	Comfort^{a}	Efficiency^{a}
Visual Flow Horizontal	Swiping left	10.77 (5.75)	2.90 (1.87)	2.36 (1.43)	2.82 (1.75)	605 (188)	1.15 (0.75)	0.89 (0.50)	0.94 (0.58)	1.18 (0.93)	1.43 (1.23)
	Swiping up	10.74 (4.93)	2.98 (1.89)	2.40 (1.79)	2.57 (1.67)	598 (195)	3.34 (2.56)	1.59 (1.30)	1.30 (1.01)	2.42 (2.08)	2.15 (1.73)
Vertical	Swiping left	11.32 (8.09)	2.87 (1.85)	2.32 (1.44)	2.75 (1.71)	567 (182)	2.93 (2.40)	1.47 (1.13)	1.44 (1.37)	2.16 (1.94)	2.08 (1.65)
	Swiping up	10.74 (5.13)	2.99 (1.89)	2.52 (1.76)	2.42 (1.49)	586 (182)	2.19 (2.16)	1.07 (0.65)	1.03 (0.59)	1.57 (1.37)	1.74 (1.41)
View		AD	BB^{c}	FCU^{c}	FDI^{a}	Rt	Intuitiveness	Learnability^{b}	Complexity	Comfort	Efficiency
Number of Entities One	Tapping twice	10.96 (5.98)	3.30 (2.34)	3.11 (1.83)	2.24 (1.43)	676 (181)	1.61 (1.53)	1.31 (0.96)	1.49 (1.17)	2.03 (1.72)	2.06 (1.94)
	Stretching	11.05 (5.76)	2.92 (1.95)	2.23 (1.23)	2.05 (1.35)	674 (176)	1.75 (1.60)	1.18 (0.83)	1.28 (1.10)	1.66 (1.38)	1.73 (1.52)
Multiple	Tapping twice	10.70 (5.72)	3.23 (2.29)	3.01 (1.75)	2.03 (1.34)	655 (226)	1.62 (1.22)	1.37 (0.82)	1.55 (1.11)	1.87 (1.26)	1.91 (1.28)
	Stretching	11.21 (6.06)	2.93 (2.00)	2.31 (1.34)	2.08 (1.38)	630 (220)	1.86 (1.25)	1.41 (1.00)	1.40 (1.07)	1.79 (1.33)	1.95 (1.47)
Remove		AD^{c}	BB	FCU	FDI^{c}	Rt	Intuitiveness^{ac}	Learnability^{a}	Complexity^{a}	Comfort^{ab}	Efficiency^{a}
Visual Flow Horizontal	Swiping left	10.75 (5.88)	2.91 (1.90)	2.58 (1.63)	2.73 (1.43)	632 (211)	4.10 (2.48)	1.89 (1.54)	1.92 (1.53)	2.89 (2.08)	3.08 (2.14)
	Swiping up	11.24 (5.51)	3.00 (1.89)	2.57 (1.87)	2.55 (1.51)	623 (192)	2.37 (1.92)	1.29 (0.93)	1.22 (0.82)	1.78 (1.45)	1.72 (1.45)
Vertical	Swiping left	10.58 (5.85)	2.94 (2.00)	2.61 (1.67)	2.83 (1.62)	614 (198)	2.94 (2.27)	1.52 (1.10)	1.31 (0.95)	2.33 (1.84)	2.22 (1.86)
	Swiping up	11.09 (5.22)	2.98 (1.75)	2.62 (2.00)	2.59 (1.51)	600 (195)	3.78 (2.62)	1.91 (1.82)	1.68 (1.28)	2.92 (2.26)	2.82 (2.20)

Table 2. The descriptive and statistical results. Muscle activities and subjective assessments are rounded to 2DP; response times are rounded to the nearest whole number, in milliseconds. Standard deviations of means are presented in parentheses.
[a] There was a statistically significant interaction between visual pattern and gesture.
[b] There was a significant main effect of visual pattern.
[c] There was a significant main effect of gesture.

4.2.1 Psychological results

For the command 'next', in terms of intuitiveness, there was a significant main effect of visual flow $[F(1,89) = 5.56, p = 0.021, \eta_p^2 = 0.059]$ and gesture $[F(1,89) = 8.591, p = 0.004, \eta_p^2 = 0.088]$. Besides, there was a significant interaction between visual flow and gesture for intuitiveness $[F(1,89) = 47.333, p < 0.001, \eta_p^2 = 0.347]$, learnability $[F(1,89) = 27.862, p < 0.001, \eta_p^2 = 0.238]$, complexity $[F(1,89) = 14.721, p < 0.001, \eta_p^2 = 0.142]$, comfort $[F(1,89) = 25.127, p < 0.001, \eta_p^2 = 0.220]$, and efficiency $[F(1,89) = 13.663, p < 0.001, \eta_p^2 = 0.133]$. Specifically, under the horizontal visual flow condition, there were significantly higher levels of intuitiveness ($p < 0.001$), learnability ($p < 0.001$), comfort ($p < 0.001$), and efficiency ($p = 0.002$), and a significantly lower level of complexity

($p = 0.001$) for swiping left with the index finger compared to swiping up; under the vertical visual flow condition, there were significantly higher levels of intuitiveness ($p = 0.046$), learnability ($p = 0.003$), and comfort ($p = 0.030$), and a significantly lower level of complexity ($p = 0.006$) for swiping up with the index finger compared to swiping left.

For the command 'view', our results demonstrated a significant main effect of number of entities for learnability [$F(1,59) = 4.895, p = 0.031, \eta_p^2 = 0.077$], showing that the level of learnability for one entity condition was significantly higher than multiple entities ($p = 0.031$).

For the command 'remove', in terms of intuitiveness, our results demonstrated a significant main effect of gesture [$F(1,89) = 5.248, p = 0.024, \eta_p^2 = 0.056$]. In terms of comfort, there was a significant main effect of visual flow [$F(1,89) = 6.242, p = 0.014, \eta_p^2 = 0.066$]. In addition, a significant interaction between visual flow and gesture was found in intuitiveness [$F(1,89) = 38.296, p < 0.001, \eta_p^2 = 0.301$], learnability [$F(1,89) = 18.902, p < 0.001, \eta_p^2 = 0.175$], complexity [$F(1,89) = 28.448, p < 0.001, \eta_p^2 = 0.242$], comfort [$F(1,89) = 23.214, p < 0.001, \eta_p^2 = 0.207$], and efficiency [$F(1,89) = 8.920, p = 0.004, \eta_p^2 = 0.091$]. Specifically, under the horizontal visual flow condition, there were significantly higher levels of intuitiveness ($p < 0.001$), learnability ($p < 0.001$), comfort ($p < 0.001$), and efficiency ($p < 0.001$), and a significantly lower level of complexity ($p < 0.001$) for swiping up with the index finger than swiping left; under the vertical visual flow condition, there were significantly higher levels of intuitiveness ($p = 0.004$) and comfort ($p = 0.033$), and a significantly lower level of complexity ($p = 0.003$) for swiping left with the index finger than swiping up. Please see Figure 9 for a visual representation of these subjective assessment results.

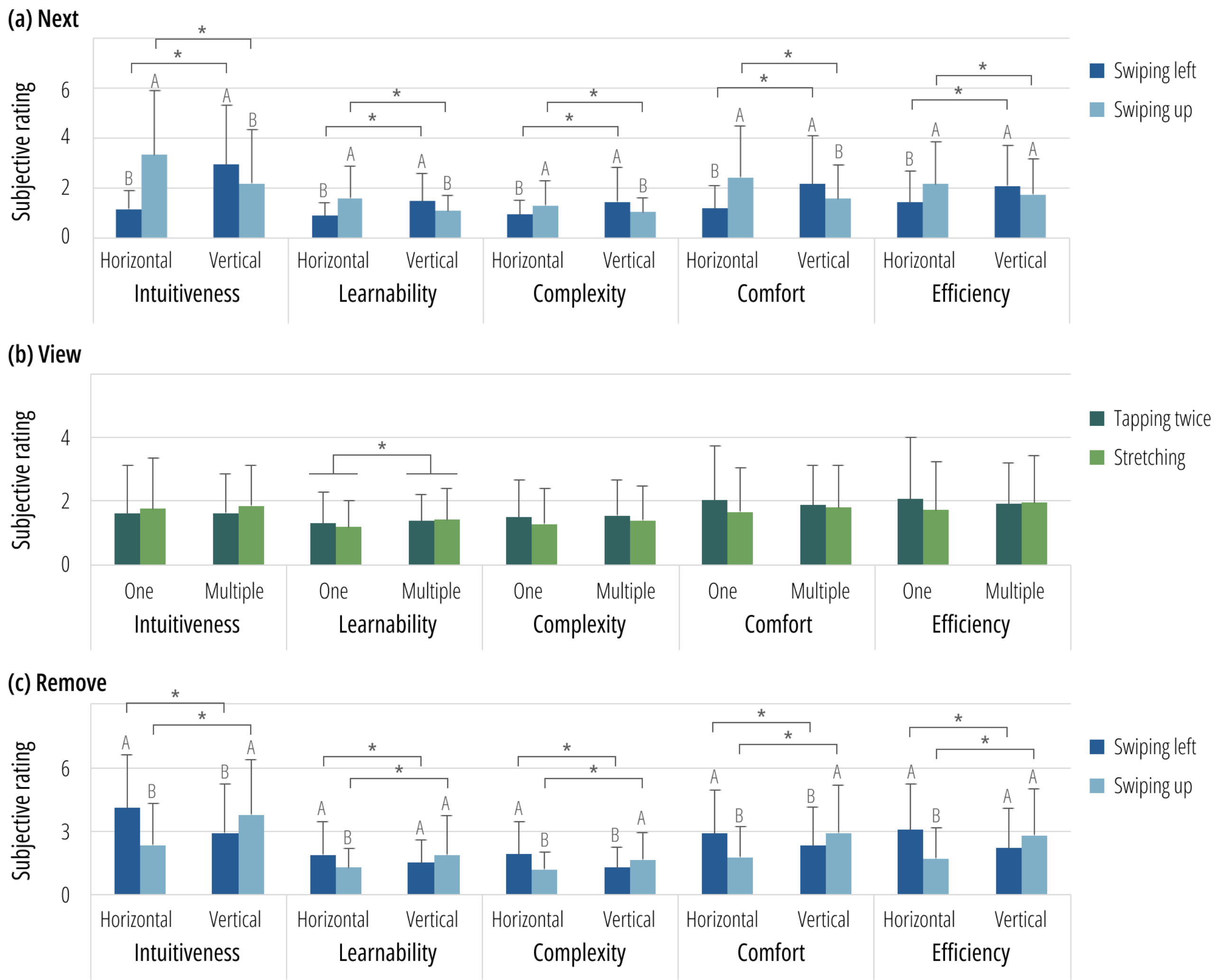

Figure 9. Results of subjective assessment. (a) The differences in the subjective ratings for the command 'next'; (b) the differences in the subjective ratings for the command 'view'; (c) the differences in the subjective ratings for the command 'remove'. * indicates $p < 0.05$.

4.2.2 Behavioural results

Trials that failed to detect a response time or where participants failed to lift their finger off the right key within the detection time range were removed from the analysis, which accounted for 8.96% of the total task response time data. For all three commands, 'next', 'view', and 'remove', there was no significant main effect or interaction between visual pattern and gesture. Please see Figure 10 for a visual representation of these task response time results.

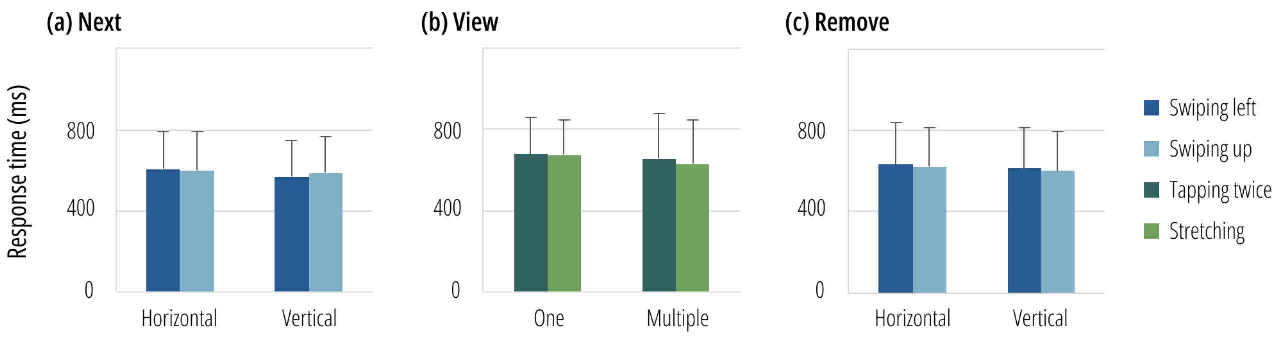

Figure 10. Results of task response time. (a) The response times for the command 'next'; (b) the response times for the command 'view'; (c) the response times for the command 'remove'.

As Kim and Lee (2023) indicated that only a part of the stimuli could show a significant difference in response times, we conducted two-sided paired t-tests on the stimuli to further explore these differences. The results of the t-tests are summarised in Table 3. For *stimulus (a)* (owing a horizontal visual flow, please refer to Figure 4), there was a significant difference in task response time between swiping left with the index finger and swiping up with the index finger ($t = -2.432, p = 0.022, Cohen'sd = -0.468$), showing that for *stimulus (a)*, the response times of performing the corresponding gesture (swiping left with the index finger) were significantly shorter than performing the non-corresponding gesture (swiping up with the index finger). For other stimuli, there was no significant difference in task response time between the corresponding and non-corresponding gestures.

	Visual Pattern	Gesture		Result			
Next Stimulus	Visual flow	Corresponding	Non-corresponding	*t*	df	*p*	Cohen's *d*
a	Horizontal	Swiping left	Swiping up	-2.432	26	**0.022**	-0.468
b	Vertical	Swiping up	Swiping left	0.555	27	0.584	0.105
c	Horizontal	Swiping left	Swiping up	1.143	25	0.264	0.224
d	Vertical	Swiping up	Swiping left	0.731	25	0.471	0.143
e	Vertical	Swiping up	Swiping left	-0.081	27	0.936	-0.015
f	Horizontal	Swiping left	Swiping up	0.787	25	0.439	0.154
View Stimuli	Number of entities	Corresponding	Non-corresponding	*t*	df	*p*	Cohen's *d*
c	One	Stretching	Tapping twice	-1.307	27	0.202	-0.247
d	One	Stretching	Tapping twice	0.993	26	0.330	0.191
e	Multiple	Tapping twice	Stretching	1.094	28	0.283	0.203
f	Multiple	Tapping twice	Stretching	-0.942	28	0.354	-0.175
Remove Stimuli	Visual flow	Corresponding	Non-corresponding	*t*	df	*p*	Cohen's *d*
a	Horizontal	Swiping up	Swiping left	0.332	25	0.743	0.065
b	Vertical	Swiping left	Swiping up	-0.753	25	0.458	-0.148
c	Horizontal	Swiping up	Swiping left	0.069	25	0.946	0.013
d	Vertical	Swiping left	Swiping up	0.153	27	0.879	0.029
e	Vertical	Swiping left	Swiping up	0.723	26	0.476	0.139
f	Horizontal	Swiping up	Swiping left	-0.771	29	0.447	-0.141

Table 3. Results for task response time across different stimuli. Significant effect is highlighted in bold. t-values, p-values and Cohen's d are rounded to 3DP.

4.2.3 Physiological results

For the command 'next', in terms of the BB, there was a significant main effect of gesture [$F(1,89) = 4.278, p = 0.042, \eta_p^2 = 0.046$], showing that swiping up with the index finger generated significantly higher muscular load than swiping left with the index finger ($p = 0.042$). In terms of the FCU, our results demonstrated a significant interaction between visual flow and gesture [$F(1,89) = 4.624, p = 0.034, \eta_p^2 = 0.049$]. Specifically, under the horizontal visual flow condition, there was no significant difference in muscle activity between swiping left and swiping up with the index finger ($p = 0.744$); however, under the vertical visual flow condition, swiping up with the index finger resulted in significantly higher muscular load compared to swiping left ($p = 0.031$). In terms of the FDI, there was a significant main effect of gesture [$F(1,89) = 7.222, p = 0.009, \eta_p^2 = 0.075$], demonstrating that swiping left with the index finger generated significantly higher muscular load than swiping up with the index finger ($p = 0.009$).

For the command 'view', in terms of the BB, there was a significant main effect of gesture [$F(1,59) = 19.701, p < 0.001, \eta_p^2 = 0.250$], showing that tapping twice with the index finger generated significantly higher muscular load than stretching with the thumb and the index finger ($p < 0.001$). In terms of the FCU, there was a significant main effect of gesture [$F(1,59) = 35.372, p < 0.001, \eta_p^2 = 0.375$], showing that tapping twice with the index finger generated significantly higher muscular load than stretching with the thumb and the index finger ($p < 0.001$). In terms of the FDI, there was a significant interaction between number of entities and gesture [$F(1,59) = 5.103, p = 0.028, \eta_p^2 = 0.080$]. Specifically, tapping twice with the index finger induced significantly greater muscular load under the one entity condition than the multiple entities condition ($p = 0.045$); however, when performing stretching with the thumb and the index finger, there was no significant difference in muscular load between the one entity condition and the multiple entities condition ($p = 0.495$).

For the command 'remove', in terms of the AD, there was a significant main effect of gesture [$F(1,89) = 6.682, p = 0.011, \eta_p^2 = 0.070$], demonstrating that swiping up with the index finger resulted in significantly higher muscular load than swiping left ($p = 0.011$). In terms of the FDI, our results demonstrated a significant main effect of gesture [$F(1,89) = 5.943, p = 0.017, \eta_p^2 = 0.063$], showing that swiping left with the index finger generated significantly higher muscular load than swiping up with the index finger ($p = 0.017$). Please see Figure 11 for a visual representation of these muscle activity results.

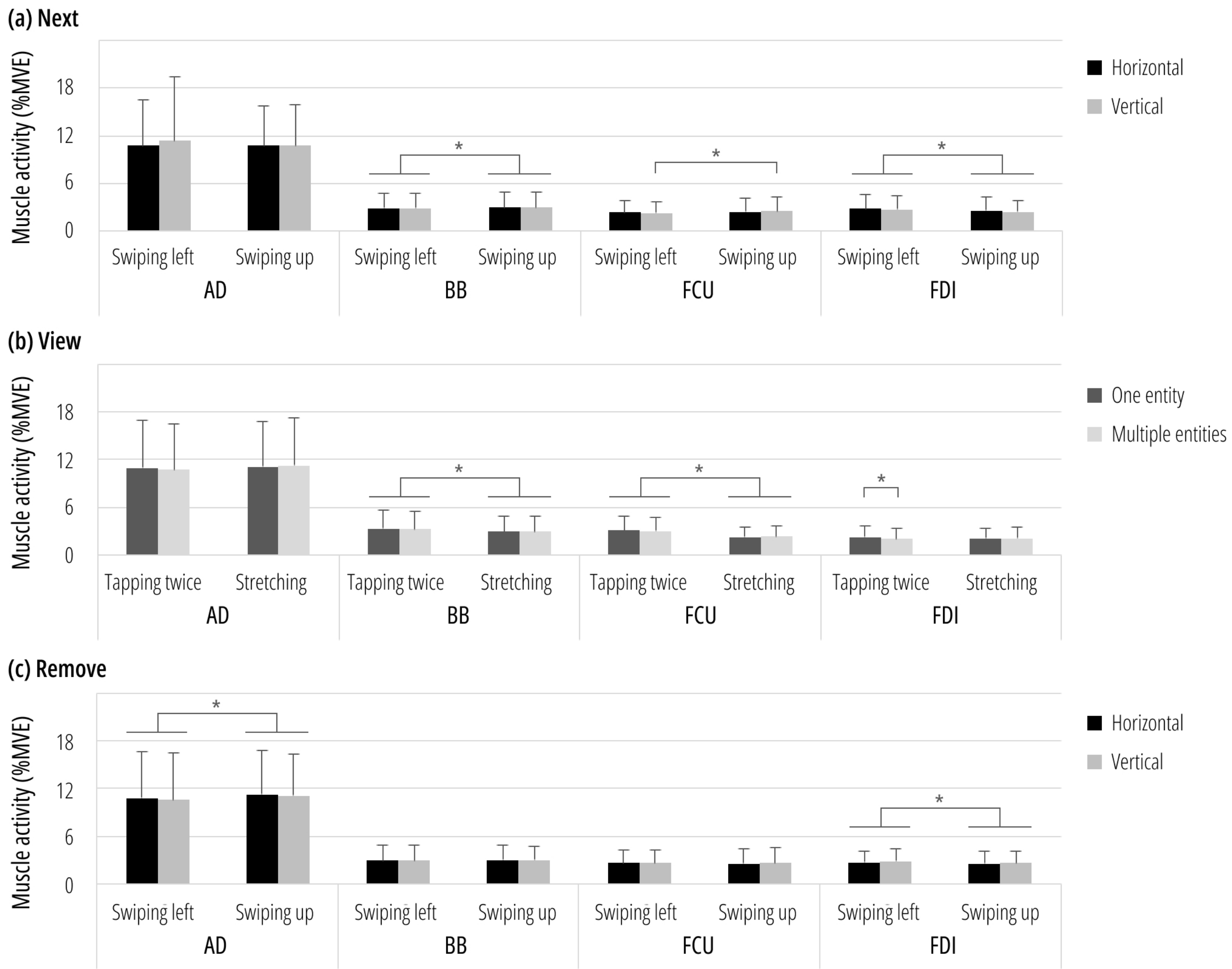

Figure 11. Results of muscle activity. (a) The differences in the muscle activities for the command 'next'; (b) the differences in the muscle activities for the command 'view'; (c) the differences in the muscle activities for the command 'remove'. * indicates $p < 0.05$.

4.3 Discussion

In experiment 2, for the three commands 'next', 'view', and 'remove', we measured the subjective assessment, task response time, and sEMG when performing corresponding and non-corresponding gestures under different visual patterns, to assess the effect of visual property-elicited mid-air gestures on human psychology, behaviour, and physiology.

4.3.1 Psychological evaluation

The subjective ratings, including intuitiveness, learnability, complexity, comfort, and efficiency were measured to assess the impact of visual patterns and mid-air gestures on human psychology.

For the command 'next', under the horizontal visual flow condition, there were higher levels of intuitiveness, learnability, comfort, and efficiency, and a significantly lower level of complexity for swiping left with the index finger compared to swiping up. According to the Gestalt principles of continuity and common fate (Wagemans et al., 2012; Wertheimer, 1923), participants might perceive the visual flow of digital entities as a unified group to be manipulated together (Kim and Lee, 2023), providing psychological advantages to mid-air gestures that aligned with the direction of visual flow for the command 'next'. Accordingly, under the vertical visual flow condition, there were higher levels of intuitiveness, learnability, and comfort, and a significantly lower level of complexity for swiping up with the index finger compared to swiping left. Generally, these subjective assessment results were in line with the results of experiment 1. Our results were partly supported by previous studies that mid-air gestures elicited by the user-led method could show high user preference and cognitive advantages (Morris et al., 2010; Wobbrock et al., 2009; Williams and Ortega, 2020). However, under the vertical visual flow condition, there was no significant difference in efficiency between swiping left and swiping up. Efficiency is often referred to as difficulty, human performance, speed, duration, and so on (Xia et al., 2022). Although swiping up might seem efficient as with the direction of the vertical visual flow, outward gestures (swiping up) generally resulted in slower task completion times compared to inward gestures (swiping left) (Coppola et al., 2018), which could offset the perceived efficiency of swiping up as the corresponding gesture, leading to no difference in the rating of efficiency under the vertical visual flow condition between swiping up and left.

For the command 'view', our results demonstrated a higher level of learnability under the one entity condition than the multiple entities condition. As learnability belongs to the cognitive factors that influence gesture interactions (Xia et al., 2022), this result might be due to the fact that the one entity condition was visually simpler than the multiple entities condition, thus reducing the mental load during the mid-air interaction.

For the command 'remove', under the horizontal visual flow condition, there were higher levels of intuitiveness, learnability, comfort, and efficiency, and a lower level of complexity for swiping up with the index finger compared to swiping left; under the vertical visual flow condition, there were higher levels of intuitiveness and comfort, and a lower level of complexity for swiping left with the index finger compared to swiping up. These results aligned with the results of experiment 1, partly supported by (Lin et al., 2021), which demonstrated that people tend to swipe perpendicular to the visual flow direction of notifications to remove them. However, there was no significant difference in learnability and efficiency between swiping left and up under the vertical visual flow condition. Learnability often refers to fast learning, memorability, recall, recognition, similarity, and fitting well with its associated function; efficiency is associated with human performance, speed, duration, and so on (Xia et al., 2022). As the unimanual pantomimic mid-air gestures for 'remove' were mostly throw-away gestures (Aigner et al., 2012), in the current study, swiping up involved a flinging motion similar to a throw-away action, effectively conveying the meaning of 'remove'. Therefore, although swiping left was perpendicular to the vertical visual flow condition, there could be no difference in the ratings of learnability and efficiency between swiping left and up.

4.3.2 Behavioural evaluation

Task response times were measured to assess the impact of visual patterns and mid-air gestures on human behaviour. For the command 'next', for stimulus (a), which featured a horizontal visual flow and multiple squares (please refer to Figure 4), the response times of swiping left with the index finger (corresponding gesture) were significantly shorter than swiping up with the index finger (non-corresponding gesture), which was consistent with Kim and Lee (2023) that for stimuli with a horizontal visual flow, the corresponding gesture resulted in significantly faster response times than the non-corresponding gesture, indicating that there was a correspondence effect on the screen, which was fundamentally derived from the correspondence between movement directions of the controlling limb and controlled elements in an operator's visual field (Hoffmann et al., 2019; Worringham and Beringer, 1998). This correspondence effect is attributed to automatic perceptuomotor processing triggered by an object's inner properties (e.g., size, orientation, shape) to invoke faster and more accurate action responses on corresponding stimuli than non-corresponding stimuli (Kim and Lee, 2020; Proctor et al., 2017; Tucker and Ellis, 1998). Therefore, our result implied a possible correspondence effect on the screen in mid-air interaction, which could evoke a direction-based correspondence effect at the user's perceptuomotor level. This result also aligned with the psychological findings that for the command 'next', under the horizontal visual flow condition, there were higher levels of intuitiveness, learnability, comfort, and efficiency and a significantly lower level of complexity for swiping left compared to swiping up with the index finger.

4.3.3 Physiological evaluation

Muscle activities of the right upper extremity muscles, including the AD, BB, FCU, and FDI, were measured to assess the impact of visual patterns and mid-air gestures on human physiology.

Next

For the command 'next', swiping up with the index finger generated significantly higher muscular load in the BB than swiping left. As the BB mainly acts on the flexion of the forearm (Jenkins, 2008), this result might be because, in the current experiment, participants tended to rotate the elbow and forearm to perform swiping up, while swiping left mainly involved finger movements. Moreover, swiping left with the index finger generated significantly higher muscular load in the FDI than swiping up. When swiping left, participants would abduct the index finger to a greater amplitude, while swiping up involved the extension of the index finger. Therefore, this result could be supported by Yahagi and Kasai (1998) that motor evoked potentials amplitudes significantly increased in both FDI muscles during motor images of flexion and abduction rather than extension.

Additionally, there was a significant interaction between visual flow and mid-air gesture regarding the muscular load in the FCU. Specifically, under the vertical visual flow condition, swiping up with the index finger resulted in significantly higher muscular load in the FCU than swiping left. Forman et al. (2019) indicated that the FCU mainly acted on flexion in the adduction direction of the wrist. Therefore, this result might be due to the fact that when participants executed swiping up with the index finger, their wrists flexed towards the ulnar side (adduction direction of the wrist); when swiping left with the index finger, their wrists flexed towards the radial side. On the contrary, under the horizontal visual flow condition, there was no significant difference in muscle activity of the FCU between swiping left and swiping up. As perceptible properties guiding a direction innate in an interface object allowed users to perceive its movement possibility in that direction (Kim and Lee, 2023), participants might perceive the possibility of horizontal movement under the horizontal visual flow condition, thus reducing the force of swiping up under the horizontal visual flow condition. Meanwhile, swiping left mainly involved wrists flexing towards the radial side and had little impact on FCU activation. Consequently, there was no difference in the muscle activity of the FCU between swiping up and left. These muscle activity results conflicted with the results of the subjective evaluation in our experiment, indicating that although swiping up as the corresponding gesture to the vertical visual flow enabled people to get a positive psychological perception, it caused greater muscular load on the forearm. Since even small movements with low forces could increase the risk of injury when repeated frequently (Asakawa et al., 2022; Gerr et al., 2014), our finding suggested that the physiological factor should be taken into consideration when designing mid-air gestures, to avoid the risk of developing musculoskeletal disorders for repetitive mid-air gesture use.

View

For the command 'view', tapping twice with the index finger generated significantly higher muscular load in the BB and FCU than stretching with the thumb and the index finger. As the BB mainly acted on the flexion of the forearm (Jenkins, 2008) and the FCU mainly acted on the flexion in the adduction direction of the wrist (Forman et al., 2019), these results probably attributed to the fact that tapping twice involved both wrist and forearm movement, while stretching with the thumb and the index finger primarily engaged finger movement without the movement of wrist and forearm.

Additionally, our results demonstrated a significant interaction between number of entities and mid-air gesture regarding the muscular load in the FDI. When stretching with the thumb and the index finger, there was no significant difference in muscle activity of the FDI between the multiple entities and the one entity condition, which might be because this stretching gesture involved spreading the fingers apart in a controlled manner, keeping the amplitude and force of this gesture similar under different conditions. When tapping twice with the index finger, there was significantly less muscular load in the FDI under the multiple entities condition than one entity condition. This result might be attributed to the limited space under the multiple entities condition, which led to lighter tapping gestures. As the FDI provides forces for the first metacarpal bone and mainly acts on and stabilises the forefinger (Putz and Pabst, 2006), the lighter gestures resulted in less muscular load in the FDI under the multiple entities condition. Although there was no significant difference in the findings from the psychological and behavioural aspects, our result suggested that tapping twice to execute 'view' was well-suited for the multiple entities condition in actual mid-air gesture interaction. This finding highlighted the importance of considering physiological load in designing mid-air gestures, even if the difference was not perceived psychologically and behaviourally.

Remove

For the command 'remove', first, swiping up with the index finger generated significantly higher muscular load in the AD compared to swiping left. This result was similar to Huang et al. (2022a) that swiping up generated greater muscular load of the AD than swiping left and right in both mid-air interaction and touchscreen interaction, indirectly supported by Asakawa et al. (2022) that swiping up demonstrated significantly greater shoulder posture angle than swiping left and right during swiping on the touchscreen. Second, swiping left with the index finger generated significantly higher muscular load in the FDI than swiping up with the index finger, which was the same as the result of the FDI for the command 'next'. As showed by Yahagi and Kasai (1998), increased motor evoked potentials amplitudes were found in FDI muscles during motor images of flexion and abduction rather than extension. Thus, this result could be because swiping left involved the abduction of the index finger and swiping up involved the extension of the index finger.

5 Conclusion

We conducted a two-stage experimental study to evaluate the effect of user-defined mid-air gestures elicited by on-screen visual properties on user experience. Initially, in experiment 1, our results determined corresponding mid-air gestures for the commands 'next', 'view', and 'remove' under specific visual patterns manipulated by visual properties. Subsequently, experiment 2 evaluated the impact of these mid-air gestures on user psychology, behaviour, and physiology. To ensure the accuracy of our analysis, we employed Create ML for mid-air gesture recognition to avoid the inclusion of incorrect or inconsistent gestures. Ultimately, the following main conclusions of this study were made:

- Experiment 1 employed a user-defined approach and choice-based gesture elicitation design to establish the corresponding mid-air gestures to different visual flow conditions for the commands 'next' and 'remove', and to different number of entities conditions for the command 'view', highlighting the role of visual property in gesture elicitation and laying the foundation for experiment 2.

- The findings of experiment 2 revealed the effect of mid-air gestures elicited by visual properties on user psychology, behaviour, and physiology.

  - Regarding visual flow, our experimental parameters could be effectively used to determine the difference in mid-air gestures under different visual flow conditions. On the psychological side, for both 'next' and 'remove' commands, under the horizontal visual flow condition, corresponding gestures exhibited higher levels of intuitiveness, learnability, comfort, and efficiency, and a lower level of complexity compared to non-corresponding gestures; under the vertical visual flow condition, corresponding gestures exhibited higher levels of intuitiveness and comfort, and a lower level of complexity compared to non-corresponding gestures, demonstrating the significant psychological benefits of well-matched gestures and visual patterns in mid-air interaction. On the behavioural side, for a stimulus consisting of multiple squares in a horizontal visual flow, executing the 'next' command by swiping left with the index finger induced significantly shorter response times compared to swiping up, suggesting a direction-based correspondence effect on the screen in mid-air interaction, impacting users at the perceptuomotor level. On the physiological side, for the command 'next', swiping up generated significantly higher forearm muscular load than swiping left under the vertical visual flow condition, suggesting that apart from the psychological factor, the physiological factor should be taken into consideration in gesture design.

  - Regarding the number of entities, we elucidated the difference in mid-air gestures between the multiple entities condition and one entity condition. We found that tapping twice with the index finger under the multiple entities condition induced less finger muscular load compared to one entity condition, highlighting the importance of considering physiological load in designing mid-air gestures, even if not perceived psychologically.

This is the first study focusing on the effect of mid-air gestures elicited by visual properties on user psychology, behaviour and physiology. The mid-air gestures identified in our experiment 1 differ from those obtained by Kim and Lee (2023) in touchscreen research. For example, in our experiment, the command 'remove' is associated with swiping left for vertical visual flow conditions and swiping up for horizontal visual flow conditions; in contrast, in the touchscreen study by Kim and Lee (2023), the majority of selections for 'remove' is long press. Moreover, our findings demonstrate that even psychologically preferred gestures can increase muscle strain. For example, although swiping left with the index finger for the 'next' command under the horizontal visual flow condition is preferred psychologically, it leads to greater muscular load than swiping up, suggesting the importance of considering physiological factors in the design of interfaces or mid-air gestures to minimise the risk of injury from frequent, repetitive motions. Our discoveries provide empirical evidence and a scientific basis for optimising gesture elicitation, enhancing user experience, and mitigating the risk of developing musculoskeletal disorders from repetitive mid-air gesture use, contributing to the development of more natural, intuitive, and ergonomically sound human-computer interactions.

There are limitations to our research that should be addressed. First, as an exploratory investigation, this study focused on four commands and considered a limited number of visual properties. Future research could expand across a broader range of commands and visual properties, ultimately developing a comprehensive set of mid-air gestures elicited by visual properties. Second, all experiments in this study were conducted in a laboratory setting. Future studies could evaluate gestures in practical scenarios, integrating system design with user psychology, behaviour, and physiology to enhance real-world applicability. In addition, this study exclusively examined 30 healthy young individuals, thereby constraining the generalisability of the sample. Considering that the physical impacts of interactions are often closely linked to age and the psychological effects may be related to professional and cultural factors, it is essential for future research to extend the scope to other demographic groups to improve the study's relevance and applicability.

6 Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendix

The statistical results of the two-way repeated measures ANOVA for the commands 'next', 'view', and 'remove' are presented in Table 4, Table 5, and Table 6, respectively.

Measure	Effect	Result				
Muscle activity		F-score	p	η_p^2	Post-hoc	95%CI
AD	Visual flow	0.516	0.474	0.006		
	Gesture	0.619	0.434	0.007		
	Visual × Gesture	0.552	0.459	0.006		
BB	Visual flow	0.091	0.764	0.001		
	Gesture	4.278	**0.042**	0.046	Left<Up	[-0.002, -0.000]
	Visual × Gesture	0.575	0.450	0.006		
FCU	Visual flow	0.725	0.397	0.008		
	Gesture	1.783	0.185	0.020		
	Visual × Gesture	4.624	**0.034**	0.049	Vertical: Left<Up	[-0.004, -0.000]
FDI	Visual flow	3.236	0.075	0.035		
	Gesture	7.222	**0.009**	0.075	Up<Left	[0.001, 0.005]
	Visual × Gesture	0.621	0.433	0.007		
Task response time		F-score	p	η_p^2	Post-hoc	95%CI
Response time	Visual flow	2.149	0.147	0.028		
	Gesture	0.199	0.657	0.003		
	Visual × Gesture	0.725	0.397	0.010		
Subjective assessment		F-score	p	η_p^2	Post-hoc	95%CI
Intuitiveness	Visual flow	5.560	**0.021**	0.059		
	Gesture	8.591	**0.004**	0.088	Horizontal: Left<Up	[-2.747, -1.627]
	Visual × Gesture	47.333	**<0.001**	0.347	Vertical: Up<Left	[0.013, 1.463]
Learnability	Visual flow	0.106	0.745	0.001		
	Gesture	3.107	0.081	0.034	Horizontal: Left<Up	[-0.991,-0.418]
	Visual × Gesture	27.862	**<0.001**	0.238	Vertical: Up<Left	[0.143, 0.650]
Complexity	Visual flow	2.748	0.101	0.030		
	Gesture	0.145	0.704	0.002	Horizontal: Left<Up	[-0.563, -0.152]
	Visual × Gesture	14.721	**<0.001**	0.142	Vertical: Up<Left	[0.122, 0.713]
Comfort	Visual flow	0.242	0.624	0.003		
	Gesture	3.529	0.064	0.038	Horizontal: Left<Up	[-1.699, -0.777]
	Visual × Gesture	25.127	**<0.001**	0.220	Vertical: Up<Left	[0.060, 1.125]
Efficiency	Visual flow	1.106	0.296	0.012		
	Gesture	1.187	0.279	0.013		
	Visual × Gesture	13.663	**<0.001**	0.133	Horizontal: Left<Up	[-1.163, -0.275]

Table 4. The statistical results of muscle activity, task response time, and subjective assessment for the command 'next'. F-scores, p-values, η_p^2, and 95% CI are rounded to 3DP. Significant effect is highlighted in bold.

Measure	Effect	Result				
Muscle activity		F-score	p	η_p^2	Post-hoc	95% CI
AD	Number of entities	0.184	0.669	0.003		
	Gesture	0.148	0.702	0.003		
	Number × Gesture	1.696	0.198	0.028		
BB	Number of entities	0.842	0.363	0.014		
	Gesture	19.701	**<0.001**	0.250	Stretching<Tapping twice	[0.002, 0.005]
	Number × Gesture	0.815	0.370	0.014		
FCU	Number of entities	0.036	0.851	0.001		
	Gesture	35.372	**<0.001**	0.375	Stretching<Tapping twice	[0.005, 0.010]
	Number × Gesture	1.580	0.214	0.026		
FDI	Number of entities	2.356	0.130	0.038		
	Gesture	0.238	0.627	0.004		
	Number × Gesture	5.103	**0.028**	0.080	Tapping twice: Multiple<One	[0.000, 0.004]
Task response time		F-score	p	η_p^2	Post-hoc	95%CI
Response time	Number of entities	3.105	0.084	0.056		
	Gesture	0.277	0.601	0.005		
	Number × Gesture	0.325	0.571	0.006		
Subjective assessment		F-score	p	η_p^2	Post-hoc	95%CI
Intuitiveness	Number of entities	0.210	0.648	0.004		
	Gesture	1.653	0.204	0.027		
	Number × Gesture	0.091	0.764	0.002		
Learnability	Number of entities	4.895	**0.031**	0.077	One<Multiple	[-0.271, -0.014]
	Gesture	0.219	0.641	0.004		
	Number × Gesture	1.884	0.175	0.031		
Complexity	Number of entities	1.236	0.271	0.021		
	Gesture	1.933	0.170	0.032		
	Number × Gesture	0.251	0.618	0.004		
Comfort	Number of entities	0.009	0.923	0.000		
	Gesture	1.426	0.237	0.024		
	Number × Gesture	1.202	0.277	0.020		
Efficiency	Number of entities	0.050	0.824	0.001		
	Gesture	0.370	0.546	0.006		
	Number × Gesture	1.771	0.188	0.029		

Table 5. The statistical results of muscle activity, task response time, and subjective assessment for the command 'view'. F-scores, p-values, η_p^2, and 95% CI are rounded to 3DP. Significant effect is highlighted in bold.

Measure	Effect	Result				
Muscle activity		F-score	p	η_p^2	Post-hoc	95%CI
AD	Visual flow	2.159	0.145	0.024		
	Gesture	6.682	**0.011**	0.070	Left<Up	[-0.009, -0.001]
	Visual × Gesture	0.026	0.873	0.000		
BB	Visual flow	0.018	0.893	0.000		
	Gesture	1.848	0.177	0.020		
	Visual × Gesture	0.795	0.375	0.009		
FCU	Visual flow	0.354	0.553	0.004		
	Gesture	0.001	0.971	0.000		
	Visual × Gesture	0.040	0.841	0.000		
FDI	Visual flow	2.030	0.158	0.022		
	Gesture	5.943	**0.017**	0.063	Up<Left	[0.000, 0.004]
	Visual × Gesture	0.453	0.503	0.005		
Task response time		F-score	p	η_p^2	Post-hoc	95%CI
Response time	Visual flow	1.821	0.181	0.024		
	Gesture	0.606	0.439	0.008		
	Visual × Gesture	0.033	0.857	0.000		
Subjective assessment		F-score	p	η_p^2	Post-hoc	95%CI
Intuitiveness	Visual flow	0.568	0.453	0.006		
	Gesture	5.248	**0.024**	0.056	Horizontal: Up<Left	[1.169, 2.280]
	Visual × Gesture	38.296	**<0.001**	0.301	Vertical: Left<Up	[-1.410, -0.272]
Learnability	Visual flow	1.479	0.227	0.016		
	Gesture	0.683	0.411	0.008		
	Visual × Gesture	18.902	**<0.001**	0.175	Horizontal: Up<Left	[0.302, 0.903]
Complexity	Visual flow	1.151	0.286	0.013		
	Gesture	3.264	0.074	0.035	Horizontal: Up<Left	[0.408, 0.983]
	Visual × Gesture	28.448	**<0.001**	0.242	Vertical: Left<Up	[-0.618, -0.129]
Comfort	Visual flow	6.242	**0.014**	0.066		
	Gesture	1.674	0.199	0.018	Horizontal: Up<Left	[0.580, 1.640]
	Visual × Gesture	23.214	**<0.001**	0.207	Vertical: Left<Up	[-1.122, -0.047]
Efficiency	Visual flow	0.666	0.417	0.007		
	Gesture	0.066	0.798	0.001		
	Visual × Gesture	8.920	**0.004**	0.091	Horizontal: Up<Left	[0.849, 1.867]

Table 6. The statistical results of muscle activity, task response time, and subjective assessment for the command 'remove'. F-scores, p-values, η_p^2, and 95% CI are rounded to 3DP. Significant effect is highlighted in bold.

Tables

Command	Visual Property	Visual Pattern	Corresponding Gesture	Non-corresponding Gesture
Next	Visual flow	Horizontal Vertical	Swiping left with the index finger Swiping up with the index finger	Swiping up with the index finger Swiping left with the index finger
View	Number of entities	One Multiple	Stretching with the thumb and the index finger Tapping twice with the index finger	Tapping twice with the index finger Stretching with the thumb and the index finger
Remove	Visual flow	Horizontal Vertical	Swiping up with the index finger Swiping left with the index finger	Swiping left with the index finger Swiping up with the index finger

Table 1. Corresponding and non-corresponding gestures to each visual pattern.

		Muscle Activity (%MVE)				Performance	Subjective Assessment				
Next		AD	BB[c]	FCU[a]	FDI[c]	Rt	Intuitiveness[a b c]	Learnability[a]	Complexity[a]	Comfort[a]	Efficiency[a]
Visual Flow											
Horizontal	Swiping left	10.77 (5.75)	2.90 (1.87)	2.36 (1.43)	2.82 (1.75)	605 (188)	1.15 (0.75)	0.89 (0.50)	0.94 (0.58)	1.18 (0.93)	1.43 (1.23)
	Swiping up	10.74 (4.93)	2.98 (1.89)	2.40 (1.79)	2.57 (1.67)	598 (195)	3.34 (2.56)	1.59 (1.30)	1.30 (1.01)	2.42 (2.08)	2.15 (1.73)
Vertical	Swiping left	11.32 (8.09)	2.87 (1.85)	2.32 (1.44)	2.75 (1.71)	567 (182)	2.93 (2.40)	1.47 (1.13)	1.44 (1.37)	2.16 (1.94)	2.08 (1.65)
	Swiping up	10.74 (5.13)	2.99 (1.89)	2.52 (1.76)	2.42 (1.49)	586 (182)	2.19 (2.16)	1.07 (0.65)	1.03 (0.59)	1.57 (1.37)	1.74 (1.41)
View		AD	BB[c]	FCU[c]	FDI[a]	Rt	Intuitiveness	Learnability[b]	Complexity	Comfort	Efficiency
Number of Entities											
One	Tapping twice	10.96 (5.98)	3.30 (2.34)	3.11 (1.83)	2.24 (1.43)	676 (181)	1.61 (1.53)	1.31 (0.96)	1.49 (1.17)	2.03 (1.72)	2.06 (1.94)
	Stretching	11.05 (5.76)	2.92 (1.95)	2.23 (1.23)	2.05 (1.35)	674 (176)	1.75 (1.60)	1.18 (0.83)	1.28 (1.10)	1.66 (1.38)	1.73 (1.52)
Multiple	Tapping twice	10.70 (5.72)	3.23 (2.29)	3.01 (1.75)	2.03 (1.34)	655 (226)	1.62 (1.22)	1.37 (0.82)	1.55 (1.11)	1.87 (1.26)	1.91 (1.28)
	Stretching	11.21 (6.06)	2.93 (2.00)	2.31 (1.34)	2.08 (1.38)	630 (220)	1.86 (1.25)	1.41 (1.00)	1.40 (1.07)	1.79 (1.33)	1.95 (1.47)
Remove		AD[c]	BB	FCU	FDI[c]	Rt	Intuitiveness[a c]	Learnability[a]	Complexity[a]	Comfort[a b]	Efficiency[a]
Visual Flow											
Horizontal	Swiping left	10.75 (5.88)	2.91 (1.90)	2.58 (1.63)	2.73 (1.43)	632 (211)	4.10 (2.48)	1.89 (1.54)	1.92 (1.53)	2.89 (2.08)	3.08 (2.14)
	Swiping up	11.24 (5.51)	3.00 (1.89)	2.57 (1.87)	2.55 (1.51)	623 (192)	2.37 (1.92)	1.29 (0.93)	1.22 (0.82)	1.78 (1.45)	1.72 (1.45)
Vertical	Swiping left	10.58 (5.85)	2.94 (2.00)	2.61 (1.67)	2.83 (1.62)	614 (198)	2.94 (2.27)	1.52 (1.10)	1.31 (0.95)	2.33 (1.84)	2.22 (1.86)
	Swiping up	11.09 (5.22)	2.98 (1.75)	2.62 (2.00)	2.59 (1.51)	600 (195)	3.78 (2.62)	1.91 (1.82)	1.68 (1.28)	2.92 (2.26)	2.82 (2.20)

Table 2. The descriptive and statistical results. Muscle activities and subjective assessments are rounded to 2DP; response times are rounded to the nearest whole number, in milliseconds. Standard deviations of means are presented in parentheses.

[a] There was a statistically significant interaction between visual pattern and gesture.

[b] There was a significant main effect of visual pattern.

[c] There was a significant main effect of gesture.

	Visual Pattern	Gesture		Result			
Next Stimulus	Visual flow	Corresponding	Non-corresponding	t	df	p	Cohen's d
a	Horizontal	Swiping left	Swiping up	-2.432	26	**0.022**	-0.468
b	Vertical	Swiping up	Swiping left	0.555	27	0.584	0.105
c	Horizontal	Swiping left	Swiping up	1.143	25	0.264	0.224
d	Vertical	Swiping up	Swiping left	0.731	25	0.471	0.143
e	Vertical	Swiping up	Swiping left	-0.081	27	0.936	-0.015
f	Horizontal	Swiping left	Swiping up	0.787	25	0.439	0.154
View Stimuli	Number of entities	Corresponding	Non-corresponding	t	df	p	Cohen's d
c	One	Stretching	Tapping twice	-1.307	27	0.202	-0.247
d	One	Stretching	Tapping twice	0.993	26	0.330	0.191
e	Multiple	Tapping twice	Stretching	1.094	28	0.283	0.203
f	Multiple	Tapping twice	Stretching	-0.942	28	0.354	-0.175
Remove Stimuli	Visual flow	Corresponding	Non-corresponding	t	df	p	Cohen's d
a	Horizontal	Swiping up	Swiping left	0.332	25	0.743	0.065
b	Vertical	Swiping left	Swiping up	-0.753	25	0.458	-0.148
c	Horizontal	Swiping up	Swiping left	0.069	25	0.946	0.013
d	Vertical	Swiping left	Swiping up	0.153	27	0.879	0.029
e	Vertical	Swiping left	Swiping up	0.723	26	0.476	0.139
f	Horizontal	Swiping up	Swiping left	-0.771	29	0.447	-0.141

Table 3. Results for task response time across different stimuli. Significant effect is highlighted in bold. t-values, p-values and Cohen's d are rounded to 3DP.

Measure	Effect	Result				
Muscle activity		F-score	p	η_p^2	Post-hoc	95%CI
AD	Visual flow	0.516	0.474	0.006		
	Gesture	0.619	0.434	0.007		
	Visual × Gesture	0.552	0.459	0.006		
BB	Visual flow	0.091	0.764	0.001		
	Gesture	4.278	**0.042**	0.046	Left<Up	[-0.002, -0.000]
	Visual × Gesture	0.575	0.450	0.006		
FCU	Visual flow	0.725	0.397	0.008		
	Gesture	1.783	0.185	0.020		
	Visual × Gesture	4.624	**0.034**	0.049	Vertical: Left<Up	[-0.004, -0.000]
FDI	Visual flow	3.236	0.075	0.035		
	Gesture	7.222	**0.009**	0.075	Up<Left	[0.001, 0.005]
	Visual × Gesture	0.621	0.433	0.007		
Task response time		F-score	p	η_p^2	Post-hoc	95%CI
Response time	Visual flow	2.149	0.147	0.028		
	Gesture	0.199	0.657	0.003		
	Visual × Gesture	0.725	0.397	0.010		
Subjective assessment		F-score	p	η_p^2	Post-hoc	95%CI
Intuitiveness	Visual flow	5.560	**0.021**	0.059		
	Gesture	8.591	**0.004**	0.088	Horizontal: Left<Up	[-2.747, -1.627]
	Visual × Gesture	47.333	**<0.001**	0.347	Vertical: Up<Left	[0.013, 1.463]
Learnability	Visual flow	0.106	0.745	0.001		
	Gesture	3.107	0.081	0.034	Horizontal: Left<Up	[-0.991,-0.418]
	Visual × Gesture	27.862	**<0.001**	0.238	Vertical: Up<Left	[0.143, 0.650]
Complexity	Visual flow	2.748	0.101	0.030		
	Gesture	0.145	0.704	0.002	Horizontal: Left<Up	[-0.563, -0.152]
	Visual × Gesture	14.721	**<0.001**	0.142	Vertical: Up<Left	[0.122, 0.713]
Comfort	Visual flow	0.242	0.624	0.003		
	Gesture	3.529	0.064	0.038	Horizontal: Left<Up	[-1.699, -0.777]
	Visual × Gesture	25.127	**<0.001**	0.220	Vertical: Up<Left	[0.060, 1.125]
Efficiency	Visual flow	1.106	0.296	0.012		
	Gesture	1.187	0.279	0.013		
	Visual × Gesture	13.663	**<0.001**	0.133	Horizontal: Left<Up	[-1.163, -0.275]

Table 4. The statistical results of muscle activity, task response time, and subjective assessment for the command 'next'. F-scores, p-values, η_p^2, and 95% CI are rounded to 3DP. Significant effect is highlighted in bold.

Measure	Effect	Result				
Muscle activity		F-score	p	η_p^2	Post-hoc	95% CI
AD	Number of entities	0.184	0.669	0.003		
	Gesture	0.148	0.702	0.003		
	Number × Gesture	1.696	0.198	0.028		
BB	Number of entities	0.842	0.363	0.014		
	Gesture	19.701	**<0.001**	0.250	Stretching<Tapping twice	[0.002, 0.005]
	Number × Gesture	0.815	0.370	0.014		
FCU	Number of entities	0.036	0.851	0.001		
	Gesture	35.372	**<0.001**	0.375	Stretching<Tapping twice	[0.005, 0.010]
	Number × Gesture	1.580	0.214	0.026		
FDI	Number of entities	2.356	0.130	0.038		
	Gesture	0.238	0.627	0.004		
	Number × Gesture	5.103	**0.028**	0.080	Tapping twice: Multiple<One	[0.000, 0.004]
Task response time		F-score	p	η_p^2	Post-hoc	95%CI
Response time	Number of entities	3.105	0.084	0.056		
	Gesture	0.277	0.601	0.005		
	Number × Gesture	0.325	0.571	0.006		
Subjective assessment		F-score	p	η_p^2	Post-hoc	95%CI
Intuitiveness	Number of entities	0.210	0.648	0.004		
	Gesture	1.653	0.204	0.027		
	Number × Gesture	0.091	0.764	0.002		
Learnability	Number of entities	4.895	**0.031**	0.077	One<Multiple	[-0.271, -0.014]
	Gesture	0.219	0.641	0.004		
	Number × Gesture	1.884	0.175	0.031		
Complexity	Number of entities	1.236	0.271	0.021		
	Gesture	1.933	0.170	0.032		
	Number × Gesture	0.251	0.618	0.004		
Comfort	Number of entities	0.009	0.923	0.000		
	Gesture	1.426	0.237	0.024		
	Number × Gesture	1.202	0.277	0.020		
Efficiency	Number of entities	0.050	0.824	0.001		
	Gesture	0.370	0.546	0.006		
	Number × Gesture	1.771	0.188	0.029		

Table 5. The statistical results of muscle activity, task response time, and subjective assessment for the command 'view'. F-scores, p-values, η_p^2, and 95% CI are rounded to 3DP. Significant effect is highlighted in bold.

Measure	Effect	Result				
Muscle activity		F-score	p	η_p^2	Post-hoc	95%CI
AD	Visual flow	2.159	0.145	0.024		
	Gesture	6.682	**0.011**	0.070	Left<Up	[-0.009, -0.001]
	Visual × Gesture	0.026	0.873	0.000		
BB	Visual flow	0.018	0.893	0.000		
	Gesture	1.848	0.177	0.020		
	Visual × Gesture	0.795	0.375	0.009		
FCU	Visual flow	0.354	0.553	0.004		
	Gesture	0.001	0.971	0.000		
	Visual × Gesture	0.040	0.841	0.000		
FDI	Visual flow	2.030	0.158	0.022		
	Gesture	5.943	**0.017**	0.063	Up<Left	[0.000, 0.004]
	Visual × Gesture	0.453	0.503	0.005		
Task response time		F-score	p	η_p^2	Post-hoc	95%CI
Response time	Visual flow	1.821	0.181	0.024		
	Gesture	0.606	0.439	0.008		
	Visual × Gesture	0.033	0.857	0.000		
Subjective assessment		F-score	p	η_p^2	Post-hoc	95%CI
Intuitiveness	Visual flow	0.568	0.453	0.006		
	Gesture	5.248	**0.024**	0.056	Horizontal: Up<Left	[1.169, 2.280]
	Visual × Gesture	38.296	**<0.001**	0.301	Vertical: Left<Up	[-1.410, -0.272]
Learnability	Visual flow	1.479	0.227	0.016		
	Gesture	0.683	0.411	0.008		
	Visual × Gesture	18.902	**<0.001**	0.175	Horizontal: Up<Left	[0.302, 0.903]
Complexity	Visual flow	1.151	0.286	0.013		
	Gesture	3.264	0.074	0.035	Horizontal: Up<Left	[0.408, 0.983]
	Visual × Gesture	28.448	**<0.001**	0.242	Vertical: Left<Up	[-0.618, -0.129]
Comfort	Visual flow	6.242	**0.014**	0.066		
	Gesture	1.674	0.199	0.018	Horizontal: Up<Left	[0.580, 1.640]
	Visual × Gesture	23.214	**<0.001**	0.207	Vertical: Left<Up	[-1.122, -0.047]
Efficiency	Visual flow	0.666	0.417	0.007		
	Gesture	0.066	0.798	0.001		
	Visual × Gesture	8.920	**0.004**	0.091	Horizontal: Up<Left	[0.849, 1.867]

Table 6. The statistical results of muscle activity, task response time, and subjective assessment for the command ‘remove’. F-scores, p-values, η_p^2, and 95% CI are rounded to 3DP. Significant effect is highlighted in bold.

Figures

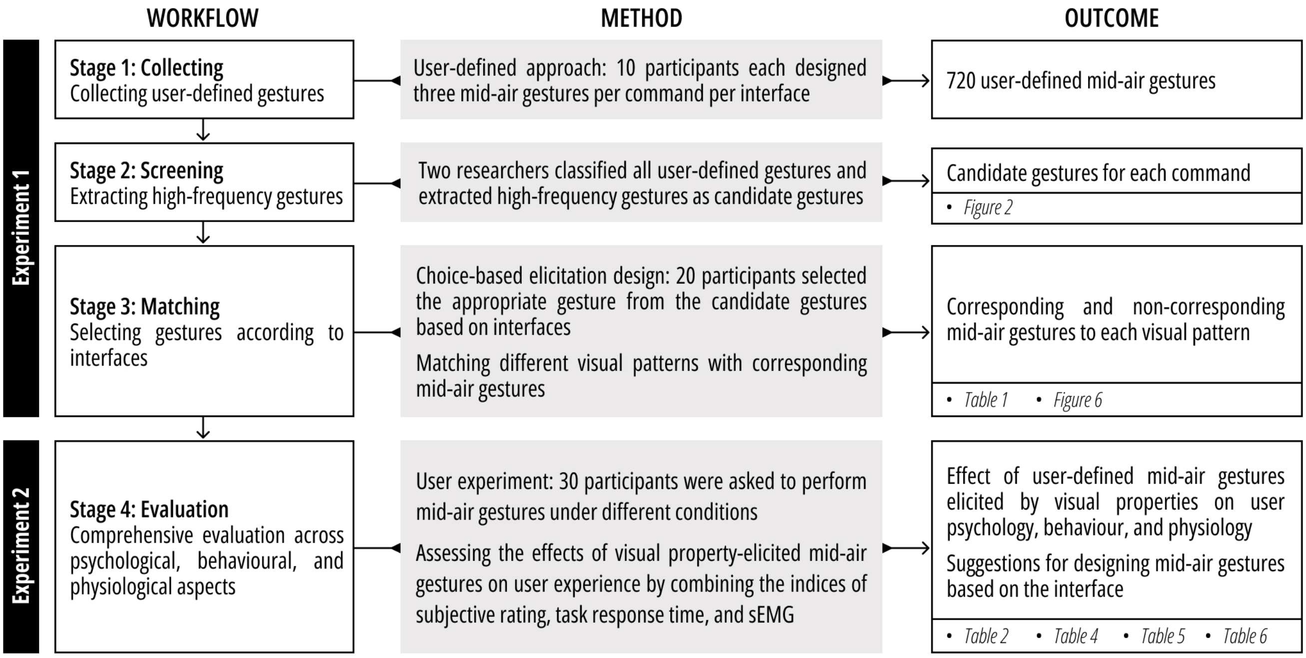

Figure 1. Workflow, methods, and outcomes of this study.

Figure 2. Illustrations of candidate gestures for the commands ‘next’, ‘select’, ‘view’, and ‘remove’.

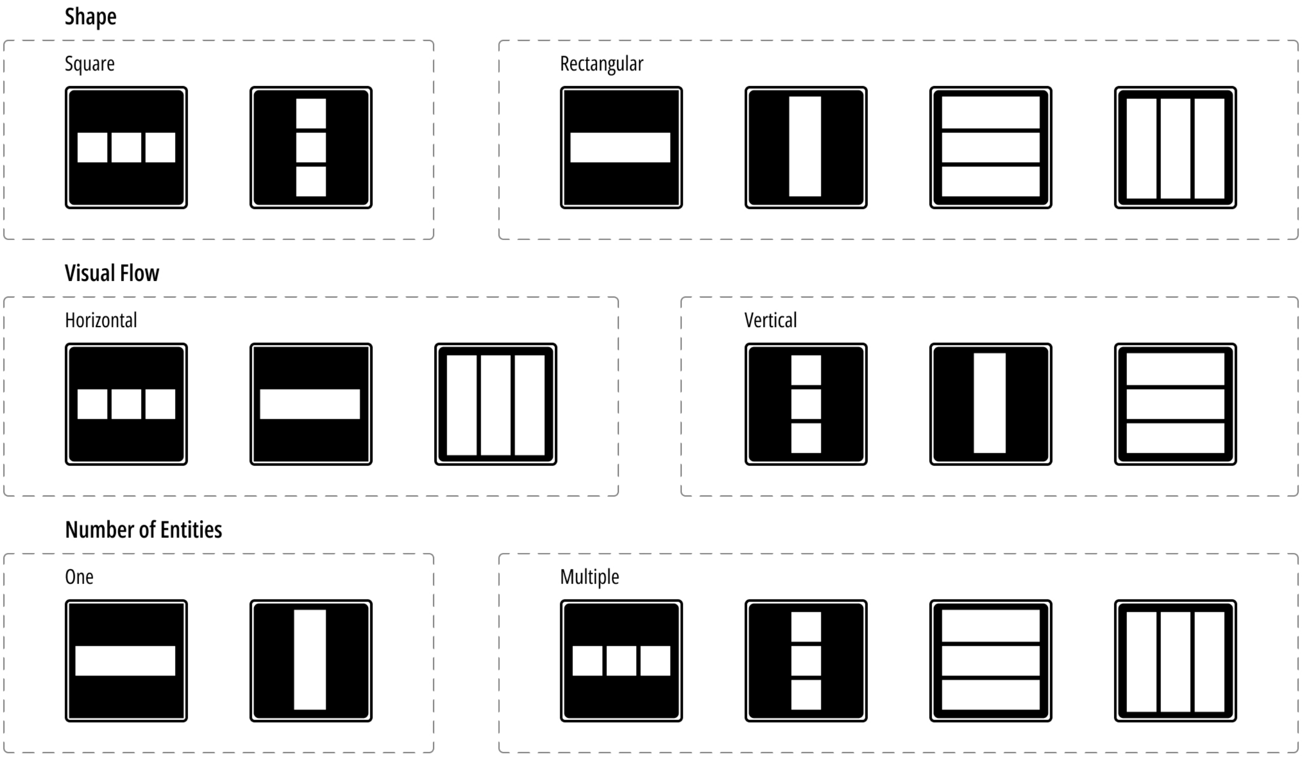

Figure 3. Interface stimulus manipulation in terms of three visual properties: shape (square and rectangular), visual flow (horizontal and vertical), and number of entities (one and multiple).

Figure 4. Full set of interface stimuli.

Figure 5. Schedule of experiment 1.

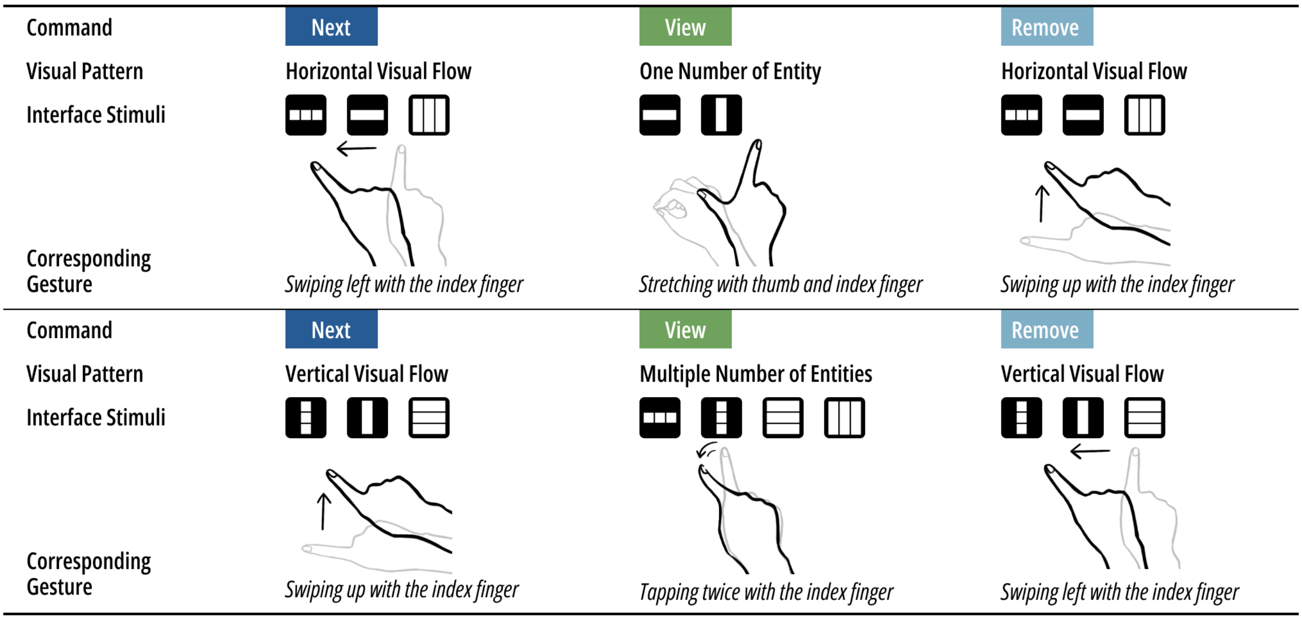

Figure 6. The corresponding gestures to different visual patterns.

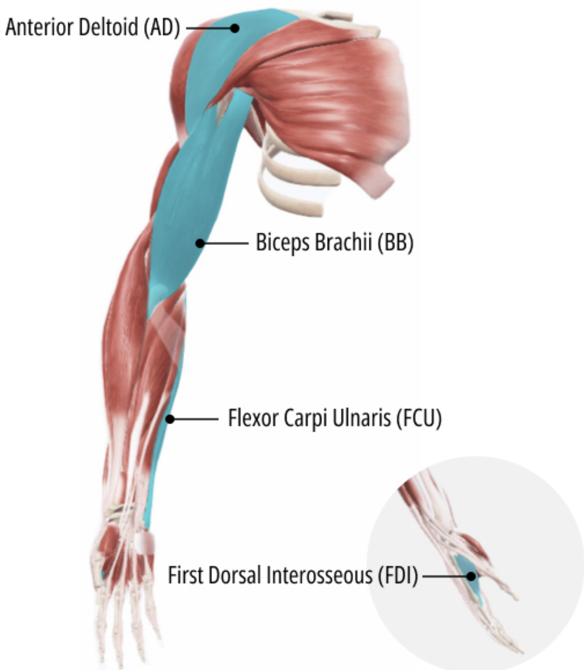

Figure 7. Illustration of four measured muscles of the upper limb.

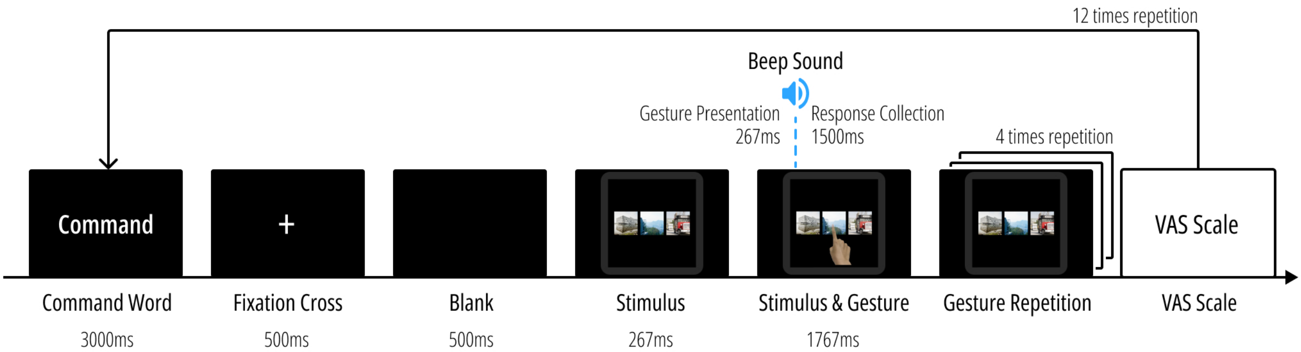

Figure 8. Schematic of experimental procedure within a block of experiment 2.

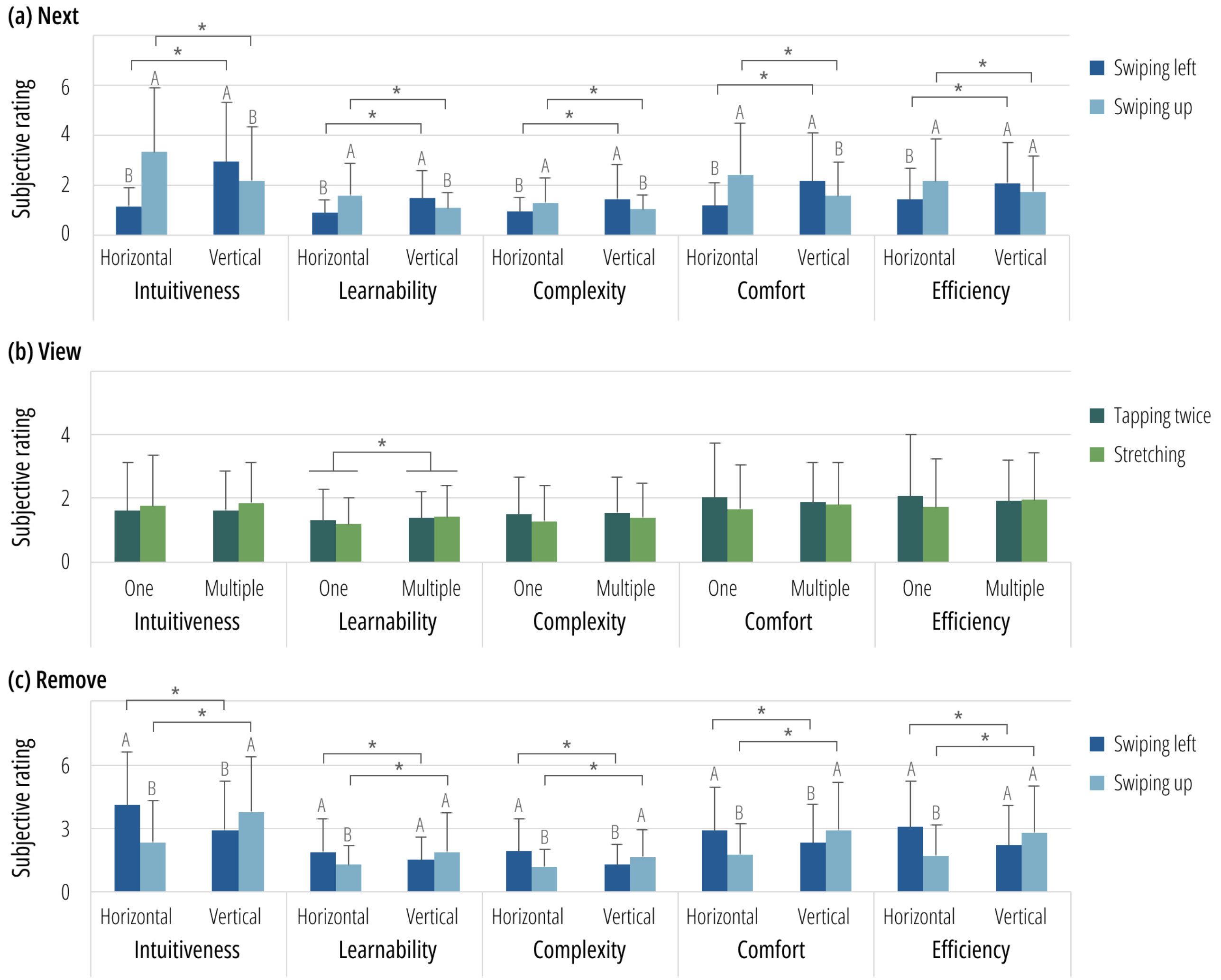

Figure 9. Results of subjective assessment. (a) The differences in the subjective ratings for the command ‘next’; (b) the differences in the subjective ratings for the command ‘view’; (c) the differences in the subjective ratings for the command ‘remove’. * indicates $p < 0.05$.

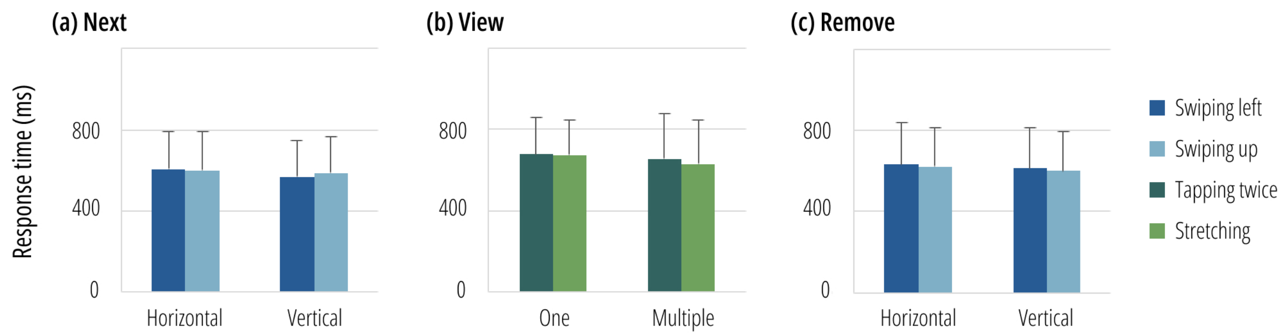

Figure 10. Results of task response time. (a) The response times for the command ‘next’; (b) the response times for the command ‘view’; (c) the response times for the command ‘remove’.

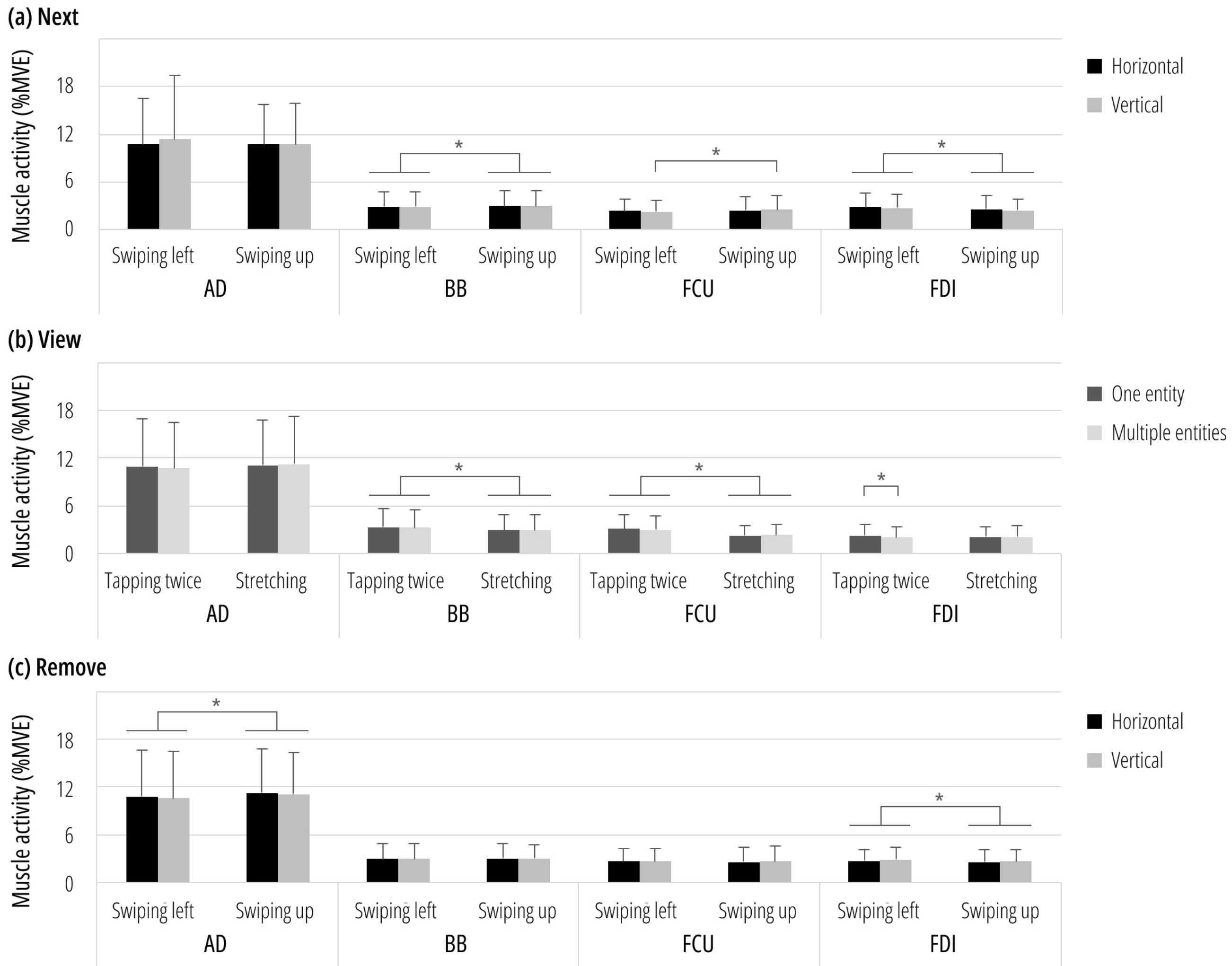

Figure 11. Results of muscle activity. (a) The differences in the muscle activities for the command 'next'; (b) the differences in the muscle activities for the command 'view'; (c) the differences in the muscle activities for the command 'remove'. * indicates $p < 0.05$.

Figure captions

Figure 1. Workflow, methods, and outcomes of this study.

Figure 2. Illustrations of candidate gestures for the commands ‘next’, ‘select’, ‘view’, and ‘remove’.

Figure 3. Interface stimulus manipulation in terms of three visual properties: shape (square and rectangular), visual flow (horizontal and vertical), and number of entities (one and multiple).

Figure 4. Full set of interface stimuli.

Figure 5. Schedule of experiment 1.

Figure 6. The corresponding gestures to different visual patterns.

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